

Course Schedule of

***‘PharmD’* Professional Degree**
Program

at the

Faculty of Pharmacy and
Pharmaceutical Sciences

Course Schedule for 1st to 5th Professional Combined.

Pharm-D. First Professional

Course No.	Course Title (First Semester)	Cr. Hours
PHT-301	Pharmaceutics- Fundamental of Pharmacy	3
PHC-303	Pharmaceutical Chemistry- Organic	3
PHL-305	Pharmacology- Islamic learning/Pak studies	2
PHL-307	Pharmacology - Physiology & Histology-I	3
PHL-309	Pharmacology - Biochemistry-I	3
PHL-311	Pharmacology - Biochemistry (Lab)	3
PHC-313	Pharmaceutical Mathematics	3
Total Courses 7		Cr. Hrs. 20

Course No.	Course Title (Second Semester)	Cr. Hours
PHT-302	Pharmaceutics-Pharmaceutical Dosage Forms-I	3
PHC-304	Pharmaceutical Chemistry -Organic/Inorganic (Lab)	3
PHC-306	Pharmaceutical Statistics	3
PHL-308	Pharmacology - Physiology & Histology -II	3
PHL-310	Pharmacology - Biochemistry -II	3
PHG-312	Pharmacognosy - General Pharmacognosy	3
PHL-314	Pharmacology – Anatomy	2
	English- Mandatory (Non-credit course)	
Total Courses 7		Cr. Hrs. 20

Pharm-D. Second Professional

Course No.	Course Title (Third Semester)	Cr. Hours
PHT-401	Pharmaceutics-Pharmaceutical Dosage Forms-II	3
PHT-403	Pharmaceutics- Microbiology- I	3
PHC-405	Pharmaceutical Chemistry-Physical-I	3
PHL-407	Pharmacology - Physiology & Histology (Lab)	3
PHL-409	Pharmacology -Pharmacology & Therapeutics-I	3
PHG-411	Pharmacognosy - Pharmacognosy -I (Lab)	3
PHL-413	Pharmacology - Pathology	2
Total Courses 7		Cr. Hrs. 20

Course No.	Course Title (Fourth Semester)	Cr. Hours
PHT-402	Pharmaceutics- Pharmaceutical Dosage Forms (Lab)	3
PHT-404	Pharmaceutics- Microbiology- II	3
PHC-406	Pharmaceutical Chemistry- Physical-I (Lab)	2
PHC-408	Pharmaceutical Chemistry- Physical-II	3
PHL-410	Pharmacology - Systemic Pharmacology	3
PHG-412	Pharmacognosy - Chemical Pharmacognosy-I	3
PHT-414	Pharmaceutics- Physical Pharmacy	3
Total Courses 7		Cr. Hrs. 20

Pharm.D. Third Professional

Course No.	Course Title (First Semester)	Cr. Hours
PHT-501	Pharmaceutics - Physical Pharmacy	3
PHC-503	Pharmaceutical Chemistry-Physical-II (Lab)	3
PHC-505	Pharmaceutical Chemistry- Quality Control	3
PHL-507	Pharmacology - Systemic Pharmacology	3
PHL-509	Pharmacology - Pathology (Lab)	2
PHG-511	Pharmacognosy - Chemical Pharmacognosy-II	3
PHT-513	Pharmaceutics - Computer Application in Pharmacy	2
Total Courses 7		Cr. Hrs. 19

Course No.	Course Title (Second Semester)	Cr. Hours
PHT-502	Pharmaceutics - Physical Pharmacy (Lab)	3
PHT-504	Pharmaceutics - Industrial Pharmacy- I (Unit operations)	3
PHC-506	Pharmaceutical Chemistry- Prep/Q.C. (Lab)	3
PHC-508	Pharmaceutical Chemistry- Pharmaceutical Analysis-I	3
PHL-510	Pharmacology - Pharmacology (Lab)	3
PHG-512	Pharmacognosy- Pharmacognosy-II (Lab)	3
PHG-514	Pharmacognosy – Natural Toxicants	2
Total Course 7		Cr. Hrs. 20

Pharm.D. Fourth Professional

Course No.	Course Title (First Semester)	Cr. Hours
PHP-601	Pharmacy Practice- I	3
PHT-603	Pharmaceutics - Industrial Pharmacy- II (Pharma. Eng.)	3
PHT-605	Pharmaceutics - Bio-Pharmaceutics & Pharmacokinetics	3
PHC-607	Pharma. Chemistry-Pharmaceutical Analysis- I (Lab)	3
PHL-609	Pharmacology - Systemic Pharmacology	3
PHG-611	Pharmacognosy- Advance Pharmacognosy	3
PHT-613	Pharmaceutics - Pharmaceutical Technology	3
Total Courses 7		Cr. Hrs. 21

Course No.	Course Title (Second Semester)	Cr. Hours
PHP-602	Pharmacy Practice -II	3
PHT-604	Pharmaceutics - Industrial Pharmacy (Lab)	3
PHT-606	Pharmaceutics - Clinical Pharmacokinetics	3
PHC-608	Pharmaceutical Chemistry- (Pharmaceutical Analysis-II	3
PHC-610	Pharmaceutical Chemistry-Medicinal-I	3
PHL-612	Pharmacology- Pharmacology (Lab)	3
PHT-614	Pharmaceutics - Pharmaceutical Technology (Lab)	3
Total Courses 7		Cr. Hrs 21

Pharm-D. Fifth Professional

Course No.	Course Title (First Semester)	Cr. Hours
PHT-701	Pharmaceutics–Biopharmaceutics & Pharmacokinetics Lab	3
PHT-703	Pharmaceutics - Forensic Pharmacy	3
PHP-705	Pharmacy Practice-III	3
PHC-707	Pharmaceutical Chemistry-Pharmaceutical Analysis-II Lab	3
PHC-709	Pharmaceutical Chemistry-Medicinal-II	3
PHL-711	Pharmacology - Clinical Pharmacology	2
PHG-713	Pharmacognosy - Clinical Pharmacognosy	2
Total Courses 7		Cr. Hrs. 19

Course No.	Course Title (Second Semester)	Cr. Hours
PHT-702	Pharmaceutics - Prescription & Community Pharmacy	3
PHT-704	Pharmaceutics – Pharma. Management & Marketing	3
PHT-706	Pharmaceutics - Prescription Pharmacy (Lab)	3
PHT-708	Pharmaceutics- Quality Control & Quality Assurance	3
PHC-710	Pharmaceutical Chemistry-Medicinal-III	3
PHL-712	Pharmacology - Toxicology	2
PHP-714	Pharmacy Practice-IV	3
Total Courses 7		Cr. Hrs. 20

Total Courses = 70

Total Credit Hours = 200

*** Passing English is mandatory**

* PHT Pharmaceutics
PHC Pharmaceutical Chemistry
PHL Pharmacology
PHG Pharmacognosy
PHP Pharmacy Practice

Course Details Department Wise from 1st to 5th PharmD Professional

Department of Pharmaceutics

Pharm-D Courses

First Year					
1st Semester			2nd Semester		
Course No.	Title of Course	Cr. Hr.	Course No.	Title of Course	Cr. Hr.
PHT-301	Pharmaceutics-Fundamentals of Pharmacy Theory	3	PHT-302	Pharmaceutics-Pharmaceutical Dosage Forms Theory (I)	3
Second Year					
PHT-401	Pharmaceutical Dosage Forms (II)	3	PHT-402	Pharmaceutical Dosage Forms (Lab)	3
PHT-403	Pharmaceutical Microbiology (I)	3	PHT-404	Pharmaceutical Microbiology (II)	3
			PHT-414	Pharmaceutics Physical Pharmacy Theory	3
Third Year					
PHT-501	Pharmaceutics-Microbiology (Lab)	3	PHT-502	Physical Pharmacy Lab	3
PHT-513	Computer Application in Pharmacy (Lab)	2	PHT-504	Industrial Pharmacy-I Pharmaceutical Unit Operations	3
Fourth Year					
PHT-603	Pharmaceutics- Industrial Pharmacy-II (Pharma. Eng.)	3	PHT-604	Pharmaceutics-Biopharmaceutics and Pharmacokinetics	3
PHT-605	Pharmaceutics-Industrial Pharmacy (Lab)	3	PHT-606	Pharmaceutics- Clinical Pharmacokinetics	3
PHT-613	Pharmaceutics-Pharmaceutical Technology	3	PHT-614	Pharmaceutics-Pharmaceutical Technology (Lab)	3
Fifth Year					
PHT-701	Pharmaceutics-Bio-Pharmaceutics & Pharmacokinetics Lab	3	PHT-702	Pharmaceutics-Prescription & Community Pharmacy	3
PHT-703	Forensic Pharmacy	3	PHT-704	Pharmaceutics-Pharmaceutical Management & Marketing	3
-	-	-	PHT-706	Prescription Pharmacy (Lab)	3
			PHT-708	Quality Control & Quality Assurance	3
Total 22 Courses making 68 Credit Hours in five years					

FIRST SEMESTER

PHT-301 Fundamentals of Pharmacy (Theory) Cr. Hrs. 3

1. Pharmacy Orientation

Introduction and orientation to the profession of pharmacy in relation to hospital pharmacy, retail pharmacy, industrial pharmacy, forensic pharmacy, pharmaceutical education and research etc.

2. History of Pharmacy

A survey of the history of pharmacy through ancient, Greek and Arab periods with special reference to contribution of Muslim scientists to pharmacy and allied sciences. Introduction to literature of pharmacy.

3. Literature of Pharmacy

Pharmacopocia, formularies, codices, abstracts, etc.

4. Physico-chemical Process

Precipitation, crystallization, evaporation, distillation, efflorescence deliquescence, lyophilization, elutriation, exication, descication, ignition, fusion, sublimation, calcination, decantation, adsorption, centrifugation, tirturation, levigation, dialysis, extraction, (maceration, percolation, infusion, decoction, digestion)

Books Recommended

1. Martin, P., Bustamante, P. and Chun, A.H.C Physical and Chemical Principles of Pharmaceutical Science, 4th Edition, New York (1999).
2. Aulton, M.E. Pharmaceutics: The Science of Dosage Form Design. 2nd Edition, Harcourt Publisher (2002).
3. Rawlins E.A. (ed.). Berdley's Textbook of Pharmaceutics, 8th (or recent edition). Macmillan Publishing Co. Inc. New York (1977).
4. Banker G.S. and Rhodes C.T. Modern Pharmaceutics, 4th Edition (Revised and expanded), Marcel Dekker, Inc. New York (2002).
5. Carstensen J.T. and Rhodes C.T. (ed.) Drug Stability: Principles and Practices, 3rd edition (revised and expanded), Mercel Dekker, New York (2000).

PHT-401 Pharmaceutical Dosage Forms (II) (Theory) Cr. Hrs. 3

Non – Galenicals – Solid Dosage Forms

1. *Powders*: Definition, properties, advantages, disadvantages, types and preparation etc.
2. *Tablets*: Definition, types, essentials, advantages, disadvantages, formulation, manufacture, evaluation etc.
3. *Capsules*: Definition, advantages, disadvantages, types etc.
4. *Miscellaneous*: Suppositories, surgical dressings, glycerogelatin, medicated pencils, cements etc.

Non-Galenical – Semi-solid Dosage Forms

1. *Ointment*: Definition, ointment bases, preparation, dispensing etc.
2. *Miscellaneous*: Creams, pastes, poultices, plasters etc.

New Dosage Forms

Introduction of new dosage forms, drug delivery system and cosmetology.

Books Recommended

1. Anya M. Hellery, Andrew W. Lloyd, James Swarbrick. Drug Delivery and Targeting. For Pharmacist and Pharmaceutical Scientist. Taylor and Francis Publications. (2001).
2. Joseph R. Robinson. Controlled Drug Delivery. Marcel Dekker Publications. (1997)
3. Ramabhadran T.V. Pharmaceutical Design and Development.
4. Aulton ME. Pharmaceutics. The Science of Dosage Forms Design.. Harcourt Publications Second Edition (2002).
5. Gilbert S. Banker. Modern Pharmaceutics, Marcel Dekker Publishing, Fourth Edition, (2002).
6. Loyd V. Allen, Jr. Howard C. Ansel. Ansel's Pharmaceutical Dosage Forms and Drug Delivery Systems. 10th Edition. Lippincott Williams E Wilkins.

PHT- 403 Pharmaceutical Microbiology (I) (Theory) Cr. Hrs. 3

1. General Microbiology

Historical introduction, scope of microbiology with special reference to pharmaceutical sciences. Nomenclature in classification of micro-organisms including virus, rickettsia, bacteria, fungi and protozoa.

2. Bacteria

Classification of bacteria, cell structure, morphology and functions, growth factors, growth characteristics and growth curve, nutritional requirement and nutritional factors effecting growth, Different type of media and staining procedures.

3. Immunology

Definitions, classifications and cellular basis of immune response, immunity, auto-immunity and tolerance. Antigen, antibody, antigen antibody reactions and their clinical and diagnostic applications. Hypersensitivity and allergy and drug allergy mechanisms. Graft reaction, autoimmune disease.

4. Infections

- a. Study of infections caused by following bacterial genera:
Staphylococcus, Streptococcus, Salmonella, Cholera, Mycobacterium, Helicobacter.
Various viral and rickettsial infections including rabies, hepatitis, AIDS and other emerging pathogens. Rickettsial infections including leishmania and protozoal infections including malaria and others. Normal flora of human body.

5. Microbiology of air, water and soil.

Books Recommended

1. Pommerville J.C., Alcamo's Fundamentals of Microbiology, Ninth Edition. Jones and Bartlett Publications.
2. Jawetz, Melnick and Aldelberg's Medical Microbiology, McGraw Hill Lange Publication, 26th Edition.

3. Greenwood D, Barer M, Slack R, Irving W. Medical Microbiology. A Guide to microbial infections. Churchill Livingstone. Eighteenth Edition.
4. Tortora. G.J, Funke B.R, Case C.L. Microbiology an Introduction. Twelfth Edition. Pearson Education.

PHT-501 Pharmaceutical Microbiology (Practical) Cr. Hrs. 3

Practical Elaborating Theory Course

1. Sterilization of glassware and pharmaceutical products by various methods.
2. Preparation of general and selective media and culturing of microorganism.
3. Total and viable counts of microorganisms, morphological and selective biochemical characteristics of some species.
4. Staining of bacteria: Gram method, acid fast, Grams staining, capsule staining, flagella and spore staining.
5. Microbiological analysis of air, water and soil.
6. Microbial evaluation of pharmaceutical products.
7. Microbiological assays of antibiotics and vitamins.
8. Pyrogen test, sterility test, toxicity test

PHT-513 Computer Application in Pharmacy (Lab) Cr. Hrs. 2

1. **Introduction** to Microsoft Windows and its different packages like MS Word, Excel, PowerPoint and Access.
2. **Internet and E-mail**
Internet and Microsoft internet explorer 5, searching the internet, e-mail and news group, favorites, security and customizing explorer.
3. **Web Page Development**
Introduction to front page, creating a first web site, basic formatting technique, manipulating tables within front page, FrontPage, pictures and multimedia, hyperlinking, bookmarks and image maps, front page and frames, managing your web, good site design.
4. **Complete Statistical Packages**, Statistica®
5. **Languages** at least two prevailing languages will be taught.

Books Recommended

Introduction to computers, Windows 7, MS office XP.

PHT-603 Industrial Pharmacy (II) Cr. Hrs. 3
Pharmaceutical Engineering (Theory)

1. **Pharmaceutical Lay Out and Plant Designing**
General layout and plant designing of the Pharmaceutical Industry Pharmaceutical plant construction, nature and properties of important materials employed in construction and erection of plant, convenience and storage of raw materials, semi finished and finished product.

2. **Design of Sterile Area**
Sterile area and its classification, ophthalmic ointments, preparation of parenterals (building, equipment), complete sterility (aseptic area), air control, (Laminar flow etc). Air locks, environmental monitoring' methods, sterilization. Filling/packaging (plastic and glass containers). Added substances (preservatives, antioxidants, solubilizer. suspending agents, buffers, stabilizers etc.)
3. **Heat Transfer**
4. **Mass Transfer**
Safety Methods in Pharmaceutical Industry
Mechanical, chemical and fire hazards problems, inflammable gases and dusts.
5. **Packaging Technology**
Influence of packaging materials, stability, packaging lines, packaging area, and packaging equipment.
6. **Equipments Used for**
Patches, sprays, implants, sutures, plasters, sachet packing.
7. **Study Tour**
A visit of the pharmaceutical industry and training will be an integral part of syllabi

Books Recommended

1. Martin, P., Bustamante, P. and Chun, A.H.C. Physical and Chemical Principles of Pharmaceutical Science, 4th Edition, New York (1999).
2. Aulton M.E. Pharmaceutics. The Science of Dosage Form Design. Harcourt Publishers. Second Edition (2002).
3. Rawlins E.A. (ed.) Berdley's Textbook of Pharmaceutics, 8th (or recent edition), Macmillan Publishing Co. Inc. New York (1977).
4. Banker G.S. and Rhodes C.T. Modern Pharmaceutics, 4th Edition (Revised and expanded), Marcel Dekker, Inc. New York (2002).
5. Carstensen J.T. and Rhodes C.T. (ed.) Drug Stability: Principles and Practices, 3rd edition (revised and expanded), Marcel Dekker, New York (2000).

PHT-605 Industrial Pharmacy (Lab) Unit Operations Cr. Hrs. 3

1. **Manufacture of Tablets by**
2. Wet granulation method, manufacture of tablets by slugging, manufacture of tablets by direct compression method.
3. **Coating of Tablets**
4. Sugar coating, film coating, enteric coating.
5. **Clarification of liquids by various processes.**
6. Size reduction, homogenization.
7. Ampoule filling, sealing and sterilization clarity and leakage tests in injectable .
8. Capsule filling by semi automatic machines.

9. Manufacture of sustained action drugs.
10. **Tablets tests:** Disintegration, dissolution, friability, hardness and thickness tests. Determination of weight variation in tablets. Density of powder. Particle size analysis.
11. Content uniformity test using HPLC technique, relative humidity and moisture content determination.

PHT-613	Pharmaceutical Technology (Theory)	Cr. Hrs. 3
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1. Barriers to Drug Delivery Systems (DDS)
 - a. Oral, IV, IM, Rectal, Pulmonary DDS
 - b. Reticulo-endothelial system
2. Principles of Non-Targeting/Conventional Pharmaceutical Formulations and Dosage Form Design
 - a. Need, product formulation and preformulation studies
 - b. Formulation development of Aerosols, Ophthalmic and Parenteral Preparations
 - c. Formulation development of Osmotic, Sustained-Release, Rapidly Disintegrating, Gastro-retentive drug delivery systems.
3. Concept of Targeting DDS
 - a. Active vs. Passive Targeting
 - b. Targeting Strategies
 - a. Environment and stimuli sensitive targeting: pH, temperature, ions, photo, magnetic, ultrasound, etc.
 - b. Regulated systems: enzyme complex etc
 - c. Ligand based targeting: Polymeric conjugates, biomacromolecular conjugates like antibody, affibody, aptamer and peptides etc
4. Introduction and Types of Various Dosage Forms and DDS
 - a. Nanoparticles, nanospheres, dendrimers, nanohydrogels, liposomes, niosomes, gold and iron oxide nanoparticles
 - b. Brief over view of theragnostic systems
5. Principles of Formulation of Novel Drug Delivery Systems
 - a. Introduction and brief overview of biomedical polymers, their physical and structure-property relationship.
 - b. Examples of bio-responsive and stimuli responsive polymers
 - c. Overview of methods to formulate polymeric nanoparticles, dendrimers, liposomes, niosomes, hydrogels and gold nanoparticles etc
 - d. Selection of ligands for delivery system with examples to targeting substrates
 - e. Formulation stability issues associated with novel DDS
6. Characterization Techniques for Novel DDS
 - a. Spectrophotometric (FT-IR, HPLC, UV, Fluorescence-spectroscopy, NMR etc)
 - b. Partcles size characterization (SEM, TEM, DLS, Size-exclusion chromatography etc)
 - c. Crystalline Structure (XRD etc)
 - d. Miscellaneous techniques (AFM, Raman-spectroscopy etc)
7. Pharmacokinetic and pharmacological aspects of novel DDS

- a. Clearance and in-vivo stability of novel DDS
 - b. Concept and strategies of designing long-circulating DDS
 - i. Phenomena of surface shape, charge, size and nature
 - ii. Effect of poly-ethylene glycol (PEG) and other long circulating polymers
 - c. Toxicity patterns of colloidal and novel drug delivery systems.
8. Drug and Disease Candidates for Novel DDS
 - a. Principles of selection of a disease
 - b. Principles of selection of drug(s) candidates for a disease
9. Commercial Benefits and Applications of Novel Drug Delivery Systems
 - a. Pharmaco-economic considerations
 - b. Advantages of novel DDS
 - c. Examples of commercialized novel DDS
10. Introduction to Pharmaceutical Bio-Technology
 - a. Introduction to basic science and application of biotechnologically derived pharmaceutical products (antibiotics, amino acids, insulin, enzymes and vaccines etc.), scope application new findings.
 - b. Role of Pharmaceutical Biotechnology in the new product development (NPD), challenges and opportunities.
11. Isolation and identification of suitable strains and their maintenance.
 - a. Genetic techniques for the strain improvement.
 - b. Effect of certain Bio-chemical parameters, media composition and process optimization
 - c. Isolation, extraction and purification techniques for the desired biopharmaceuticals.
 - d. Upstream and downstream processes.
 - e. Area design and quality requirements during manufacturing of biopharmaceutical.
 - f. Commercial implications of recombinant DNA technology.
 - g. The concept of gene therapy to cure genetic diseases.
12. Pharmaceutical Biotechnology and Health with special reference to treatment of cancer and possibly AIDS.

Books Recommended

1. Gilbert S. Banker, Modern Pharmaceutics, Marcel and Dekker Publishing (20020
2. Pharmaceutical Biotechnology: Fundamentals and Applications – October 22, 2013 by Daan J. A. Crommelin (Editor), Robert D. Sindelar (Editor), Bernd Meibohm (Editor).
3. Pharmaceutical Biotechnology: Drug Discovery and Clinical Applications – May 21, 2012, by Oliver Kayser (Editor), Heribert Warzecha (Editor).
4. Pharmaceutical Biotechnology: Concepts and Applications [Kindle Edition], Gary Walsh (Author).
5. Pharmaceutical Biotechnology: Concepts and Applications [Kindle Edition], Gary Walsh (Author).
6. “Drug Delivery: Engineering Principles for Drug Therapy” Edited by W. Mark Saltzman, Oxford University Press, Montreal Canada. (2001)

7. “Cellular Drug Delivery: Principles and Applications” Edited by D. Robert Lu and Svein Øie, Humana Press, Springer Science, New York, USA. (2004)
8. “Biomaterials Science: An Introduction to Materials in Medicine”, Third Edition by Buddy A. Ratner, Allan S. Hoffman, Frederick J. Schoen and Jack E. Lemons, Academic Press, Elsevier, Amsterdam, The Netherlands. (2013)
9. “Nanochemistry” by G. B. Sergeev, Elsevier Science, Amsterdam, The Netherlands. (2006)
10. Recent Review and tutorial articles published in high impact journals like Nature and Nature sister journals

PHT-701

**Pharmaceutics-Biopharmaceutics &
Pharmacokinetics Lab**

Cr. Hrs. 3

1. To classify 4 Model drugs into respective classes of “Biopharmaceutics Classification System” according to FDA guidelines: (5 Practical to be performed)
 - a. Construction of **standard curve** of 4 model drugs by UV-visible spectroscopy
 - b. Construction of **pH-Solubility** profile of 4 model drugs at **pH 1.2, 4.5 and 6.8**
 - c. Calculation of **pKa** of 4 model drugs by **Handerson-Hasselbalch** equation
 - d. Determination of **logP** values by **shake flask method** and calculation of **logD (pH 1.2, 4.5 and 6.8)** values of 4 model drugs
 - e. Determination of “**dissolution rate**” of 4 model drugs in 30 mins by USP method I and II

Activity after 5 experiments

Finally students should discuss the results of 5 experiments and predict the Biopharmaceutics classes of the drugs their absorption site in GIT and formulation priority for the respective drugs.

2. To estimate pharmacokinetic parameters using plasma drug concentration vs time curve on semi-log graphs (manually) and comparison with parametric values obtained through data plotting and estimation by PK solver. **[Each practical shall involve manual as well as software based PK calculations and then comparative assessment of parameters obtained from two methods]** **(Total of 8 Experiments)**
 - a. **IV bolus administration**
 - i. Non-compartment
 - ii. One-compartment
 - iii. Two-compartments
 - iv. Three-compartments
 - b. **Oral administration**
 - i. Non-compartment
 - ii. One compartment
 - c. **IV infusion**

d. Urinary Excretion Data

3. To estimate oral pharmacokinetic parameters using plasma drug concentration vs time curve after oral administration by **Wagner-Nelson method**.
4. To perform **IVIVC** determination of **2 model drugs data** (dissolution and in-vivo)
 - a. By Convolution method
 - b. By De-convolution method
5. To prepare **plasma samples** for HPLC based analysis of 2 model drugs (high soluble and low soluble).
6. Introduction and performance on **WinNonLin** using PK data of 2 model drugs
7. To determine '**plasma protein binding**' of **2 model drugs** by
 - a. **Gravimetric Method**
 - b. **Spectrophotometric Method**
 - c. **Other Relevant Method**
8. To perform **Clinical Pharmacokinetics** based "**Therapeutic Drug Monitoring**" of models drugs according to established relevant clinical guidelines
 - a. **Manual calculations based studies.**
 - b. **Software based (JKPD) development of "user defined model" and application**
 - i. Vancomycin
 - ii. Gentamicin
 - iii. Cyclosporin
 - iv. Digoxin
 - v. Phenytoin
 - vi. Theophylline
 - vii. Carbamazepine

Books Recommended

1. Bauer, L. 2008 Applied Clinical Pharmacokinetics, 2nd ed., McGraw-Hill, ISBN #0071476288
2. Birkett, D.J. 2002. Pharmacokinetics Made Easy, McGraw-Hill Australia, North Ryde NSW Australia, ISBN 0 074 71072 9
3. Bonate, P.L. 2011 Pharmacokinetic-Pharmacodynamic Modeling and Simulation, 2nd ed, Springer, New York, NY ISBN-13 978-1441994844
4. Bonate, P.L. and Howard, D.R., ed. 2004 Pharmacokinetics in Drug Development: Regulatory and Development Paradigms, Volume 2, AAPS Press, Arlington, VA ISBN 0-9711767-3-6
5. Boroujerdi. M. 2002 Pharmacokinetics: Principles and Applications, McGraw-Hill Professional Publishing, ISBN: 0-07135-164-7
6. Bourne, D.W.A. 1995 Mathematical Modeling of Pharmacokinetic Data, Technomic Publishing Company, Lancaster, PA ISBN 1-56676-204-9
7. Burton, M.E., Shaw, L.M., Schentag, J.J., and Evans, W.E. (editors) 2006. Applied Pharmacokinetics & Pharmacodynamics, Principles of Therapeutic Drug Monitoring, Lippincott Williams & Wilkins, Baltimore, MD ISBN 0-7817-4431-8
8. Carstensen, J.T. 1996. Modeling and Data Treatment in the Pharmaceutical Sciences, Technomic Publishing Co., Inc., Lancaster, PA ISBN 1-56676-440-8

9. Crommelin, D. J. A. and Sindelar, R. D., 2007 Pharmaceutical Biotechnology, 3rd ed., Taylor & Francis, Philadelphia, PA ISBN 0415285011
10. Crommelin, D. J. A., Sindelar, R. D., Meibohm, B., 2007 Pharmaceutical Biotechnology:
11. D.Argneio, D.Z. 2004 Advanced Methods of Pharmacokietic and Pharmacodynamic Systems Analysis Vol.3, Kluwer Academic Publishers, Norwell, MA ISBN 1-4020-7804-8
12. Florence A.T. and Attwood, D. 2006 Physicochemical Principles of Pharmacy, 4th ed., Pharmaceuitcal Press, London, UK. ISBN 0 85369 608 X
13. Gabrielsson, J and Hjorth, S. 2012 Quantitative Pharmacology, An Introduction to Integrative Pharmacokinetic-Pharmacodynamic Analysis Swedish Pharmaceutical Press, Stockholm, Sweden ISBN 9789197945233
14. Gabrielsson, J. and Weiner, D. 2007 Pharmacokinetic and Pharmacodynamic Data Analysis, Concepts and Applications, 4th ed., CRC Press, Baco Raton, FL ISBN 978-9-1976-5100-4
15. Gibaldi. M. and Perrier, D. 1982 Pharmacokinetics, 2nd edition, Marcel Dekker, New York. ISBN 0-8247-1042-8-6
16. Hedaya, M.A. 2007 Basic Pharmacokinetics, CRC Press, Baco Raton, FL ISBN 978-1-4200-4671-7
17. Humma, L.M., Ellingrod, V.L. and Kolesar, J.M. 2003 Lexi-Comp's Pharmacogenomics Handbook, Lexi-Comp, Hudson, OH ISBN 1-59195-060-0
18. Kallen, A. 2007 Computational Pharmacokinetics, CRC Press, Baco Raton, FL ISBN 978-1-4200-6065-2
19. Karch, S.B. 2007 Pharmacokinetics and Pharmacodynamics of Abused Drugs, CRC Press, Boca Raton, FL ISBN 978-1-4200-5458-3
20. Kimko, H. and Duffull, S.B. 2002 Simulation for Designing Clinical Trials, A Pharmacokinetic-Pharmacodynamic Modeling Perspective, Marcel Dekker, New York, NY ISBN 0-8247-0862-8
21. Krishna, R. 2004 Applications of Pharmacokinetic Principles in Drug Development, Kluwer Academic/Plenum Publishers, New York, NY ISBN 0-306-47766-1
22. Krishna R. 2006 Dose Optimization in Drug Development Taylor and Francis/Informa Healthcare ISBN: 1574448080
23. Krishna, R. and Yu, L., 2008 Biopharmaceutics Applications in Drug Development, Springer ISBN: 978-0-387-72378-5
24. Kwon, Y. 2001 Handbook of Essential Pharmacokinetics, Pharmacodynamics, and Drug Metabolism for Industrial Scientists, Kluwer Academic/Plenum Publishers, New York, NY ISBN 0-306-46234-6
25. Meibohm, B., 2006 Pharmacokinetics and Pharmacodynamics of Biotech Drugs: Principles and Case Studies in Drug Development, Wiley-VCH, Weinheim, Germany. ISBN 3-527-31408-3
26. Murphy J.E. 2001 Clinical Pharmacokinetics, 2nd ed., American Society of Health-System Pharmacists, Bethesda, MD ISBN 1-879907-98-4
27. Notari, R.E. 1987 Biopharmaceutics and Clinical Pharmacokinetics, 4th edition, Dekker. New York, NY ISBN 0-8247-7523-6
28. Peters, S.A. 2012 Physiologically-Based Pharmacokinetic (PBPK) Modeling and Simulations Principles, Methods, and Applications in the Pharmaceutical Industry, Wiley, ISBN 978-0-470-48406-7
29. Reddy, M.B., Yang, R.S.H., Clewell, H.J. and Andersen, M.E. 2005 Physiologically Based Pharmacokinetic Modeling, Wiley, Hoboken, NJ ISBN 0-471-47814-8
30. Ritschel, W.A. and Kearns, G.L. 2004 Handbook of Basic Pharmacokinetics ... including Clinical Applications, 6th ed., American Pharmaceutical Association, Washington, DC ISBN 1-58212-054-4
31. Rowe, P. 2012 Pharmacokinetics, Free at bookboon.com ISBN 978-87-403-0090-1

32. Schoenwald, R.D. 2002 Pharmacokinetics in Drug Discovery and Development, CRC Press, Boca Raton, FL ISBN 1-56676-973-6
33. Semla, T.P., Beizer, J.L. and Higbee, M.D. 2003 Lexi-Comp's Geriatric Dosage Handbook, 9th ed., Lexi-Comp, Hudson, OH ISBN 1-59195-067-8
34. Shargel, L., Wu-Pong, S. and Yu, A.B.C. 2012 Applied Biopharmaceutics and Pharmacokinetics, 6th ed., McGraw-Hill, New York, NY ISBN 978-0-07-160393-5
35. Testa, B., van de Waterbeemd, H., Folkers, G., and Guy, R. (ed) 2001 Pharmacokinetic Optimization in Drug Research, Verlag Helvetica Chimica Acta, Zurich, ISBN 3-906390-225
36. Tozer, T.N. and Rowland, M. 2006 Introduction to Pharmacokinetics and Pharmacodynamic: The Quantitative Basis of Drug Therapy, Lippincott Williams & Wilkins, Baltimore, MD ISBN 0-7817-5149-7
37. Wagner, J.G. 1975 Fundamentals of Clinical Pharmacokinetics, Drug Intelligence
38. Wagner, J.G. 1993. Pharmacokinetics for the Pharmaceutical Scientist, Technomic Publishing Company, Lancaster, PA ISBN 1-56676-032-1
39. Washington, N., Washington C. and Wilson, C.G. 2001 Physiological Pharmaceutics, Barriers to Drug Absorption, Taylor and Francis, Inc., New York, NY ISBN 0-748-40562-3
40. Winter, M.E. 2004 Basic Clinical Pharmacokinetics, 4th ed., Lippincott Williams & Wilkins, Baltimore, MD ISBN 0-7817-4147-5

PHT-703

Forensic Pharmacy

Cr. Hrs. 3

1. Study of Drug Laws
The Drugs Act 1976 and rules framed there under. Provincial drug rules (Respective Drug Rules will be taught in the relevant province). Advertisement rules. Other related rules and Legal aspects.
2. The Pharmacy Act 1967 and Amendate Pharmacy Act of 1971
3. The Dangerous Drugs Act, 1930
4. The Factory Law 1934
5. Shops and Establishment Ordinance, 1969 with rules
6. The Poisons Act 1919
7. Control of narcotics substances Act 1997

Books Recommended

1. The Manual of Drug Laws in Pakistan.
2. The Factory Law (1934).
3. Shop and Establishment Ordinance (1969).
4. Control of Narcotics Substances Act (1997).

SECOND SEMESTER

PHT-302 *Pharmaceutical Dosage Forms Theory (I)*

Cr. Hr. 3

1. Classification of Pharmaceutical Dosage Forms

Significance of classification, different classifications based on the following; methods of preparation. Galenicals and non-galenicals preparations, sterile and non-sterile preparations. Physical classification, pharmacological classification, classification according to mode of

application, Pharmaceutical classification according to release of drug, conventional dosage forms, time release products, repeat action products, prolonged action products, sustained released products, classification according to name of manufactures.

2. **Pharmaceutical Dosage Forms**

Galenical liquid aqueous dosage forms

Infusions, decoctions, galenical liquid non-aqueous dosage forms, alcoholic dosage forms (tinctures, fluidextracts), oleagenous dosage forms (oleoresins, infused oils), galenical solid dosage forms. Extracts, resins.

Non galenical aqueous liquid dosage forms

Aqueous dosage forms, waters, diluted acids solutions, douches, enemas, gargles, washes, juices, ophthalmic solutions. Sweet viscid aqueous dosage forms, syrups, honeys, mucilages, jellies, aqueous suspensions, suspensions, mixtures, magmas, gels, lotions.

Non-glaenical non-aqueous dosage forms

Alcoholic solutions, spirits, elixirs, dental liniments, etherial solutions (collodions), glycerine solution (glycerites), oleagenous dosage forms (oleovitamins, toothache drops, inhalations, sprays, liniments, emulsions.

3. **Parenterals**

Definition, history, types, advantages, disadvantages uses, parenteral production area, containers and closures vehicles, pyrogens, requirements, manufacturing, testing etc.

Books Recommended

1. Aulton M.E. *Pharmaceutics. The Science of Dosage Form Design*. Harcourt Publishers. Second Edition (2002).
2. Rawlins E.A. (ed.). *Bentley's Textbook of Pharmaceutics*, 8th (or recent edition). Macmillan Publishing Co. Inc. New York (1985).
3. Alfonso R Gennaro. *Remington The Science and Practice of Pharmacy* 20th Edition. Mack Publishing Company. (2002).
4. Winfield J. and Richards RME. *Pharmaceutical Practice*. Second Edition. Prentice Hall Publishing. (2004).

PHT-402

Pharmaceutical Dosage Forms (Lab)

Cr. Hrs. 3

Practical Elaborating Theory Course

Preparation of simple syrups, emulsions. Ointments, lotions, throat paints, glycerites, linimints, sprits, poultices, granules.

Books Recommended

1. Aulton M.E. *Pharmaceutics. The Science of Dosage Form Design*. Harcourt Publishers. Second Edition (2002).
2. Rawlins E.A. (ed.). *Bentley's Textbook of Pharmaceutics*, 8th (or recent edition). Macmillan Publishing Co. Inc. New York (1985).

PHT-404 Pharmaceutical Microbiology (II) (Theory) Cr. Hrs. 3

1. **Sterilization and Disinfection**
Methods of sterilization and disinfection, mechanism of killing. Kinetics of death of micro-organism, evaluation of sterilization and disinfection procedures.
2. **Fermentation**
Fermentation and respiration, ranges of fermentation, parameters of fermentation. Production of pharmacologically active fermentation products, selection of a suitable medium, isolation and recovery, kinetics of microbial growth, continuous culture application in fermentation, engineering aspect of fermentation.
3. **Biologicals**
Vaccines, classification of vaccines, production of vaccines and antisera. DNA recombinant technology for preparing vaccines Hormones and other biologicals. Pyrogen, sterility and toxicity tests.
4. **Antibiotics**
Definitions, classification of antibiotics, mode of actions, antimicrobial spectrum and side effects of different groups of antibiotics.
5. **Factory and Hospital Hygiene**
Introduction, control of microbial contamination during manufacturing, manufacturing of sterile products.

Books Recommended

1. Hugo W.B. and Russell A.D. Pharmaceutical Microbiology.
2. Presscot and Dunn Industrial Microbiology.
3. Daan JA Crommelin, Robert D Sindelar. Pharmaceutical Bio-technology. An Introduction for Pharmacist and Pharmaceutical Scientist. Harwood Academic Publishers. (1997).
4. Pelczar and Chan Microbiology.
5. Meckallee M. Microbiology Essentials and Applications by Collen and Lynes. Microbiology Methods.

PHT-414 Physical Pharmacy Cr. Hrs. 3

1. **Physico-chemical Principles**
Solutions: Introduction, types, concentration expressions ideal and real solution, colligative properties, their mathematical derivations and application in Pharmacy, molecular weight determinations, distribution co-efficient and its applications in Pharmacy solubilization, solubility factor affecting solubility. HLB-value, surfactants, their properties and types. Micelles, their formulation and types, ionization. pH, pH indicators, pK_a, buffer's equation. Isotonic solutions and their applications in pharmacy.
2. **Micromeritics**
Particle size and shapes, distribution of particles methods of determination of particle size and importance of particle size in pharmacy.
3. **Dispersion**
Colloids: Types, methods of preparation, properties (optional, kinetic, electrical). Dialysis and artificial kidney, stability of colloids, protection and sensitization phenomenon and application of colloids in Pharmacy.
Emulsions: Types, theories of emulsification, emulsifying agents their classification and stability of emulsion.
Suspensions: Types, methods of preparation, properties, suspending agents, their classification and stability.
Adsorption: Techniques and processes of adsorption in detail.

4. **Rheology**

Definition and fundamental concept. Properties contributing to rheological behaviour. Graphic presentation of rheological data.

5. **Rate and order of reactions.**

6. Kinetic Principles and Stability Testing, Theoretical Considerations

Degradation, physical factors, influence of pH, temperature, ionic strength, acid base catalysis, UV light. Chemical factors. Complex chemical reactions, oxidation-reduction hydrolysis.

Books Recommended

1. Martin, P., Bustamante, P. and Chun, A.H.C. Physical and Chemical Principles of Pharmaceutical Science, 4th Edition; New York (1999).
2. Aulton M.E. Pharmaceutics. The Science of Dosage Form Design. Harcourt Publishers. Second Edition (2002).
3. Rawlins E.A. (ed.) Bentley's Textbook of Pharmaceutics, 8th (or recent edition). Macmillan Publishing Co. Inc. New York (1977).
4. Banker G.S. and Rhodes C.T. Modern Pharmaceutics, 4th Edition (Revised and expanded), Marcel Dekker, Inc. New York (2002).
5. Carstensen J.T. and Rhodes C.T. (ed.) Drug Stability: Principles and Practices, 3rd edition (revised and expanded), Marcel Dekker, New York (2000).

PHT-502

Physical Pharmacy (Lab)

Cr. Hrs. 3

Practical Elaborating Theory Course

Experiments to demonstrate some of physico-chemical processes, like simple distillation, steam distillation, crystallization, Dialysis.

Determination of emulsion systems. Practicals based on rheological and structural character of emulsions stabilized by mixed films of emulsifier.

Determination of particle size, angle of repose of powders. Preparation of buffer solutions and isotonic solution.

Determination of percentage composition of solutions by specific gravity method.

Partition-coefficient, surface tensions, viscosity.

Determination of various pH by acidic and alkaline buffers.

Drug stability experiments, preparation of stock solution (dilution method).

Determination of critical micelle concentration (CMC) of a surface active agents. Flocculation and deflocculation of Kaolin Suspensions.

Books Recommended

1. Martin, P., Bustamante, P. and Chun, A.H.C. Physical and Chemical Principles of Pharmaceutical Science, 4th Edition, New York (1999).
2. Aulton M.E. Pharmaceutics. The Science of Dosage Form Design. Harcourt Publishers. Second Edition (2002).
3. Rawlins E.A. (ed.) Bentley's Textbook of Pharmaceutics, 8th (or recent edition), Macmillan Publishing Co. Inc. New York (1977).
4. Banker G.S. and Rhodes C.T. Modern Pharmaceutics, 4th Edition (Revised and expanded), Marcel Dekker, Inc. New York (2002).
5. Carstensen J.T. and Rhodes C.T. (ed.) Drug Stability: Principles and Practices, 3rd edition (revised and expanded), Marcel Dekker, New York (2000).

PHT-504 Industrial Pharmacy (I) Pharmaceutical Unit Cr. Hrs. 3
Operations (Theory)

Mixing

Fundamentals, mechanism. Mixing equipment used in liquid/liquid, liquid/solid and solid/solid mixing. Comminution (size reduction), Reasons for size reduction. Factors affecting size reduction, size analysis. Sieving, energy mills (Ball mill, edrunner, edge runner mill disintegrant, colloid mill, hammer mill, cutter mill, fluid energy mill etc.).

Drying

Theories of drying, drying of solids, classification of dryers, general methods, fluidized bed systems, pneumatic systems, spray dryer, freeze drying

Clarification and Filtration

Theory, filler media, filter aids, filter selection, equipment used for filtration.

Evaporation

General principles of evaporation, evaporators, evaporation under reduced pressure.

Compression and Compaction

The solid-air interface, angle of repose, flow rates, mass volume relationship, density, Heckel plots, consolidation, granulation, friability, compression (dry method, wet method, slugging), physics of tableting. Tableting machines and other equipment required, problems involved in tableting, tablet coating.

Encapsulation

Capsulation hard and soft gelatin capsules.

Books Recommended

1. Aulton M.E. *Pharmaceutics. The Science of Dosage Form Design*. Harcourt Publishers. Second Edition (2002).
2. Leon Lachman, Lea and Febiger. *Theory and Practice of Industrial Pharmacy* (1986).
3. James Swarbrick. *Encyclopedia of Pharmaceutical Technology*, Marcel and Dekker Publishing (2002).

PHT-604 Bio-Pharmaceutics and Pharmacokinetics (I) Cr. Hrs. 3
(Theory)

1. Biopharmaceutics

Definition and concept, bioavailability, physiology of the G.I.T. Physiologic factor affecting the physiochemical. Formulation factor. Dissolution, factor effecting the dissolution for the absorption of the drug. Physiology of the skin. Physiological factor effect the absorption of the drug through the skin. Parenteral release dosage form. Route of administration, physiological factor effecting the absorption of the drug. Bio-pharmaceutics of other dosage form.

2. Pharmacokinetics

Review of mechanical fundamentals, rate and order of reaction. Definition and concept of pharmacokinetics. Determination through plasma level.

Books Recommended

1. Leon Shargel. *Biopharmaceutics and Pharmacokinetics*. Fourth Edition (2002).
2. Ronald D. Schoenwald. *Pharmacokinetics Principles of Dosing Adjustment*. (2002).
3. Milo Gibaldi. *Biopharmaceutics and Clinical Pharmacokinetics*. Fourth Edition (1991).

PHT-606

Clinical Pharmacokinetics (II)

Cr. Hrs. 3

Compartment models. One compartment model. Two compartment models. Three compartment models. Non-compartmental models.

Biological half-life in vitamin.

Clearance, elevation.

Protein bounding.

Multiple dosage

Application of pharmacokinetics in clinical situation.

Application in dosage sites.

Bioavailability and bioequivalence testing.

Pharmacokinetics of intravenous infusion.

PHT-614

Pharmaceutical Technology (Lab)

Cr. Hrs. 3

1. To prepare matrix tablet of drugs by single-punch machine
 - a. Preparation of matrix tablet for less water soluble drug
 - b. Preparation of matrix tablet for highly water soluble drug
 - c. Perform dissolution testing by USP method I and II
2. Formulation of 2 Model Drugs (highly soluble and less soluble) in Nano particles/Micro particles
 - a. To prepare polymeric Nano/Micro particles by single / double emulsion / nanoprecipitation method(s)
 - i. **Hydrophilic** drugs by **double emulsion method**
 - ii. **Hydrophobic** drugs by **single emulsion method**
 - iii. **Hydrophobic** drugs by **nanoprecipitation method**
 - b. To visualize the nano/microparticles under stereomicroscope
 - c. To perform the release testing of 2 model drugs from nanoparticles stirring method (UV-visible spectroscopy)
3. To prepare the oral formulation of drugs by extrusion/spheronization method
 - a. Matrix based pellets
 - b. Coated pellets (perform coating)
 - c. Pelltization to form immediate release, controlled/extended release and enteric coated pellets.
4. To observe the geometry and surface of spheronized formulation (pellets) by stereomicroscope
5. To perform the release testing of drugs from spheronized (pellets) particles by USP method I and II
6. To compare and study the swelling and disintegration profile (physical deformation) of expired vs. valid film, enteric coated tablets and hard gelatin capsules
[the experiment involves the physical phenomena that happens to expired tablets specially the coating of polymers since they tend to be oxidized or cured over the period of time and may hamper the disintegration of tablets and capsules.
7. To develop transdermal patches of suitable active pharmaceutical excipients.
8. To develop bilayer-tablet formulations of different active pharmaceutical ingredients.
9. Fast dispersible tablet formulations different NSAIDs.
10. To develop osmotically-controlled tablet formulations for control over extended period of time.
11. To preparation pharmaceutical hydrogels
12. To prepare liposomal formulations of 2 model drugs (highly soluble and low soluble)

PHT-702 Prescription Pharmacy and Community Pharmacy Cr. Hrs. 3

1. **Basic Principles of Compounding and Dispensing**
Weights and measures. Calculations for compounding and dispensing. Fundamental preparations in compounding. Containers and closures for different products. Prescription handing, Parts of prescription, filling, labelling, pricing of dispensed medication.
2. **Extemporaneous Dispensing**
3. **Pharmaceutical Incompatibilities**
Types of incompatibilities. Manifestations, correction and prevention with reference to typical examples.
4. I. V. admixtures
5. Dispensing of radio pharmaceuticals
6. Definitions and background of community pharmacy
7. Epidemiology and its control. Prevents Health (EPI & CDC). Family planning. Health policy and National drug policy.
8. Patient assessment.
9. Medical compilation of drug administration (General & Socio-economic Aspects).
10. Patient pharmacist communication.
11. Patient education and counseling.
12. Control of drug abuse and misuse.
13. Role of pharmacist as Public Health Educator in the community for drug monitoring and drug information

Books Recommended

1. Carter SJ. Cooper and Guns. Dispensing for Pharmaceutical Students (1997).
2. Roy Robertson. Management of Drug Users in the Community: A practical Handbook edited.
3. Alfonso R Gennaro. Remington The Science and Practice of Pharmacy 20th Edition. Mack Publishing Company (2002).
4. Martindale's Extra Pharmacopoeia. 32nd edition. (2002).

PHT-704 Pharmaceutical Management and Marketing Cr. Hrs.3
(Theory)

Management

Nature and principles of management, types and functions of managers.

Planning

Purpose and types of planning, steps in planning. Organizing, management control systems. Purpose, steps in the control process. Forms of operations control. Requirements for adequate control. Critical control points and standards. Motivation, innovation and creativity, communication.

Production Management

Material management, supply chain, demand management, regulatory control.

Marketing Management

Marketing channels, promotion, advertising and salesmanship. Promotion marketing.

Sales Management

Personnel, buying, receiving and pricing, Sales promotion and customer services.

Pharmacy Layout Design

Objectives of layout design. Types of community pharmacies. Pharmaceutical centre, prescription-oriented pharmacies, traditional pharmacies, the super drug store.

Consumer goods and purchases. Classes of layout designs. Principles and characteristics of layout design. Traffic flow analysis.

Books Recommended

1. Patrick Tharp C. and Pedro J. Lecca. Pharmacy Management for students and practitioners.
2. Harry A. Smith. Principles and Methods of Pharmacy Management.
3. Dalton E. McFarland. Management – Foundations and Practices.
4. Kreitner. Management.
5. Newman and Summer. The Process of Management.

PHT-706

Prescription Pharmacy (Lab)

Cr. Hrs. 3

Practicals Elaborating Theory Course

Preparation and dispensing of:

Syrups, Mixtures, Ointments, Creams, Lotions, Pastes, Ear drops, Glycerite, Mouth wash, Nasal drops etc.

PHT-708

Pharmaceutical Quality Control and Quality

Cr. Hrs. 3

Assurance (Theory)

1. Validation of Pharmaceutical Process

Control of components and drug product containers and doses. Production and process controls. Packaging and labelling controls. Holding and distribution. Repackaging and relabelling. Regulating basis for person validation, sterilization validation of sterile products. Sterile product validation. Validation of solid dosage form. Process validation and quality assurance. Prospective process validation, validation for water system for sterile and non-sterile products, cleaning validation, equipment validation, process validation of raw materials. Analytical method validation, computer system validation, validation of diluted aerosole

2. **Different Quality Control tests** of liquids, emulsion, solid state and time release product. Biological assays. Biological methods. Bioassay of antibiotics. Standard preparation and unit of activity. Assay of vitamins, hormones miscellaneous tests. Toxicity tests and identification test, ash contents etc. General knowledge of B.P, B.P.C. USP, N. etc. Statistical interpretation of quality controls data. Quality control and assurance of hospital and clinical pharmacy.

Books Recommended

1. Sydney H. Willig. Good Manufacturing Practices for Pharmaceuticals. Marcel Dekker Publishing.
2. Pharmaceutical Process Validation. Marcel Dekker Publishing (1999)
3. Joseph. T. Dipiro. Encyclopedia of Pharmaceutical Technology. Marcel Dekker Publishing (2003).

Department of Pharmaceutical Chemistry

Pharm-D Courses

First Year					
1st Semester			2nd Semester		
Course No.	Title of Course	Cr. Hr.	Course No.	Title of Course	Cr. Hr.
PHC-303	Organic Chemistry	3	PHC-304	Organic Chemistry (practical)	3
PHC-313	Pharmaceutical Mathematics	3	PHC-306	Pharmaceutical Statistics	3
			-	-	-
Second Year					
PHC-405	Physical Chemistry I	3	PHC-406	Physical Chemistry I (Practical)	2
-	-	-	PHC-408	Physical Chemistry II	3
Third Year					
PHC-503	Physical Chemistry II (Practical)	3	PHC-506	Preparation And Quality Control (Practical)	3
PHC-505	Theoretical Basis of Quality Control	3	PHC-508	Pharmaceutical Analysis-1	3
Fourth Year					
PHC-607	Pharmaceutical Analysis-1 (Practical)	3	PHC-608	Pharmaceutical Analysis-1I	3
-	-	-	PHC-610	Medicinal Chemistry-I	3
Fifth Year					
PHC-707	Pharmaceutical Analysis-1I (Practical)	3	PHC-710	Medicinal Chemistry-III	3
PHC-709	Medicinal Chemistry-II	3	-	-	-
Total 17 Courses making 50 Credit Hours in five years.					

FIRST SEMESTER

First Professional

PHC-303
Organic Chemistry
Cr. Hrs. 3

1. Introduction to pharmaceutical chemistry, classification and nomenclature of organic pharmaceutical compounds. hyperconjugation, steric effects inductive effect and mesomeric effect,.
2. Nucleophilic and electrophilic substitution reaction in aromatic system. Theory of resonance.
3. Orientation in electrophilic substitution reactions on benzene ring.
4. Organic reactions: Baeyer-Villiger oxidation; Diels Alder reaction; Grignard's reaction, metal hydride reduction and Wolf Krishner reduction, Friedel Craft's reaction, Perkin reaction, Cannizzaro reaction.
5. Carbonium ion rearrangements; Pinacol-pinacolone, Wagner-Merrwein, Wolf, Hofmann and Beckmann rearrangements.
6. Carbanions; condensation reaction (Aldol condensation; Favorskiirearrangment; Wittig reaction).
7. Stereoisomerism, optical isomerism, geometrical isomerism, tautomerism of carbonyl compounds, resolution of racemic mixture and conformational analysis.
8. Free radicals: Introduction, structure and stability.
9. General methods of preparations, properties, identification test and pharmaceutical applications of the following classes and their analogs:
alcohols, phenols, ethers, aldehydes, ketones, carboxylic acids, esters, amines and aniline.

Books Recommended

1. A Guide Book to Mechanisms in Organic Chemistry Peter Sykes. Fourth edition, Longman Group Ltd. (1978).
2. Introduction to Organic Chemistry By William H. Brown Saunders College Publishing 3rd edition (2002).
3. Organic Chemistry Graham Solomons T.W. Fifth edition, John Wiley & Sons, Inc. (1992).
4. Textbook of Organic Medicinal and Pharmaceutical Chemistry Wilson and Gisvold's. 10th edition, Lippincott-Raven Publisher (1998).
5. The Organic Chemistry of Drug Design and Drug Action, Richard B. Silver. Academic Press, Pnen, Inc., USA (1992).

PHC-313
Pharmaceutical Mathematics
Cr. Hrs. 3

1. Algebra
Common and natural logarithm, solution of linear and quadratic equations. Equations reducible to Quadratic form, solution of simultaneous equations. Arithmetic, Geometric and harmonic progressions. Arithmetic, geometric and harmonic means, permutations and combinations. Binomial Theorem: Simple application.
2. **Trigonometry**

Measurement of angles in radian and degrees. Definitions of circular functions. Derivation of circular function for simple cases and trigonometry identities.

3. Analytical Geometry

Coordinates of point in a plane. Distance between two points in a plane. Locus. Equations of straight parallel and perpendicular lines. Equation of parabola, circle and ellipse.

4. Differential Calculus

Limit, concept of derivative. Rules of differentiation, examples on the evaluation of derivatives. Derivatives of algebraic, trigonometric, exponential and logarithmic functions, partial differentiation, higher order derivatives. Maxima and minima points of inflections.

5. Integral Calculus

Concept of integration. Rules of integration's. Integration's of algebraic, exponential, logarithmic and trigonometric functions by using different techniques and numerical integration.

Books Recommended

1. Edwards, C.H. Jr. and David E. Penney. Calculus and Analytical Geometry (1995).
2. Text Book of Algebra and Trigonometry for Class XI and XII (New edition).

Second Professional

PHC-405

Physical Chemistry I

Cr. Hrs. 3

1. Physical Properties and Molecular Constitution

Surface and interfacial tension, dielectric constant, dipole moment, refractive index, optical rotation, density, specific gravity, viscosity, molar refraction, parachor.

2. Colloids and Colloidal System

Characteristics features of colloids, type of colloidal system, properties of colloids and colloidal systems. Preparation and purification of colloidal solutions, stability of colloids, pharmaceutical applications.

3. Solutions

Definition types and properties, concentration, solubility and solubilization, factors affecting solubility, solvents used in pharmacy, solutions of electrolytes and non-electrolytes, isotonic solutions, dissolution and dissolution rates, distribution phenomena, theory of distillation, molecular weight determination.

4. Ionic Equilibria

Modern theory of acids, bases and salts, acid-base equilibria, pH and acidity constants, theory of indicators, buffers and buffered system.

5. Phase Equilibria

Phase rule and its applications to one and two component system.

Books Recommended

1. British Pharmacopoeia, Stationary Press, Royal Society of Pharmaceutical Press, London (2005).
2. Physical Pharmacy Martir A.N., 4th Edition, Lea and Febiger, Philadelphia (1993).
3. Practical Pharmaceutical Chemistry Backet and Stenlake., Royal Press, London (1998).
4. Remington Practical of the Science and Pharmacy, Mack Publishing Company, Eston, Pennsylvania, USA.
5. United State Pharmacopoeia, United State Pharmacopoeial Convention, Inc., 12601. Twinbrook Parkway, Rockyville M.D. 20852 USA. 2008

Third Professional

PHC-503

Physical Chemistry II (Practical)

Cr. Hrs. 3

1. Conductometric determination of the ionization constant of a weak acid.
2. Conductometric determination of the solubility of sparingly soluble salt.
3. U. V. absorption spectra of aromatic compounds and geometric isomers.
4. Determination of the wavelength of maximum absorbance and molar extinction coefficient of a given sample.
5. Determination of the first-order rate constant for acid catalysed hydrolysis of a given sample.
6. Determination of the first-order rate constant for the decomposition of a given sample.
7. Determination of effect of change of temperature in the rate of reaction.
8. To study the stability of a drug subjected to various stress conditions
9. Determination of heat of neutralization of HCl and NaOH.
10. Determination of heat of vapourization / transition of a given sample.
11. Determination of heat of solution from solubility.
12. Determination of an equilibrium constant by distribution method.

Books Recommended

1. Pharmaceutical Analysis, Higuchi T. and BrochmannHanssen E., Interscience, New York.
2. Physical Chemistry Barrow G.M... McGraw-Hill, London (1973).
3. Physical chemistry by Fazal I Hussain, 3rd Edition (1991).
4. Physical Chemistry, Barrow G.M... McGraw-Hill, London (1973).
5. Text Book of Physical Chemistry by SameulGlasstone 2nd Edition MC Millan& Company Ltd. (1946).

PHC-505

Theoretical Basis of Quality Control

Cr. Hrs. 3

1. Pharmaceutical chemistry and its relation to other sciences.
2. Brief historical outline.
3. Raw material for drugs
4. Sources of impurities in pharmaceuticals.
5. Purity and its control.
6. Testing and determination of water.
7. Survey of analytical methods
8. Quantitative and qualitative analysis (general information)
9. Pharmacopoeial tests and specifications.
10. Standardization of pharmaceuticals and formulated products.
11. Quality control system for drugs and pharmaceuticals
12. Causes of poor quality and general requirements.
13. Total quality management, a new approach.
14. Sampling techniques, validation and statistical treatment of analytical data.

Books Recommended

1. Pharmaceutical Analysis – A Textbook of Pharmacy, Student and Pharmaceutical Chemistry Watson D.G. Churchill Livingston (1999).
2. Quantitative Analysis, Theory and Practice by Lawrence W. Pott Harper and raw Publisher New York (1987).
3. Remington Practical of the Science and Pharmacy, Mack Publishing Company, Eston, Pennsylvania, USA.
4. United State Pharmacopoeia, United State Pharmacopoeial Convention, Inc., 12601. Twinbrook Parkway, Rockyville M.D. 20852 USA. 2008
5. Vogel's Text Book of Quantitative Organic Chemistry, Longman, London. 6th Edition (2006).

Fourth Professional

PHC-607

Pharmaceutical Analysis – I (Practical)

Cr. Hrs. 3

1. **Assay of pharmaceutical compounds based on chemical methods**
Acid base titration, oxidation-reduction titration, complexometric titration, gravimetric, solvent extraction, and gasometric methods.
Potentiometric determination
pH of a solution and titration of an acid
Strength of unknown solution of HCl with NaOH.
Strength of each acid in a mixture of HCl and CH₃COOH using standard alkali.
Conductometric determination
Equivalent point of titration of HCl with NaOH

2. **Polarographic determination**
Amount of nitrobenzene in solutions.
Quinhydrone.
3. **Determination of refractive index of liquids by Abbe-refractometer.**
Determination of the composition of unknown mixture of two liquids
Determination of the concentration of alcohol in solutions.
Determination of a 10% solution of calcium chloride.
4. **Polarometric determination of:**
Concentration of sugar in a given solution.
Percentage of two optically active substances in a given solution.
5. Fluorimetric determination of a standard drug in a given sample.

Books Recommended

1. British Pharmacopoeia, Stationary Press, Royal Society of Pharmaceutical Press, London
2. United State Pharmacopoeia, United State Pharmacopoeial Convention, Inc., 12601. Twinbrook Parkway, Rockyville M.D. 20852 USA. 2008
3. Practical Pharmaceutical Chemistry Backet and Stenlake., Royal Press, London (1998).
4. Quantitative Analysis, Theory and Practice by Lawrence W. Pott Harper and raw Publisher New York (1987).
5. Vogel's Textbook of Quantitative Inorganic Analysis including Elementary Instrumental Analysis, Longman, 6th Edition, London (2006).

Fifth Professional

PHC-707 Pharmaceutical Analysis – II (Practical) Cr. Hrs. 3

1. Assay of pharmaceutical compounds based on spectrophotometric methods.
2. Separation, identification and quantitation of a drug substance by chromatographic methods such as TLC, GLC, HPLC.
3. To study the methods development procedure of a drug substance by UV-Visible spectroscopy/TLC/GLC/HPLC.
4. To study the experimental methodology of validation of a drug substance by UV-Visible spectroscopy/TLC/GLC/HPLC.

Books Recommended

1. British Pharmacopoeia, Stationary Press, Royal Society of Pharmaceutical Press, London (2005).
2. Instrumental Methods of Chemical Analysis by Golden W. Evings, McGraw Hill London 5th Edition (1985).
3. Principle of Instrumental Analysis, Skoog Holter and Nieman 5th Edition (2000).
4. United State Pharmacopoeia, United State Pharmacopoeial Convention, Inc., 12601. Twinbrook Parkway, Rockyville M.D. 20852 USA. 2008
5. Vogel's Text Book of Quantitative Organic Chemistry, Longman, London. 6th Edition (2006).

PHC-709**Medicinal Chemistry-II****Cr. Hrs. 3**

To study the chemistry, structure, mechanism of action, structure activity relationship and therapeutic applications of the following:

1. Analgesic and Antipyretics

Paracetamol, salicylic acid analogues, quinolines derivatives pyrazolone and pyrazolodines, N-arylanthranilic acids, aryl and heteroaryl acetic acid derivatives.

2. Local Anaesthetics

Benzoic acid derivatives, lidocaine derivatives (anilides), amino benzoic acid, miscellaneous compounds such as: Procaine, lignocaine, eucaine, cocaine and benzocaine.

3. Central Nervous System Depressants

General anesthetics, inhalation anesthetics, ultrashort acting barbiturates, dissociative anesthetics such as cyclopropane, halothane, nitrous oxide, chloroform, thiopental sodium, ketamine, methohexital, thioamylal sodium, fentanyl citrate, tribromo ethanol.

Anxiolytics, sedative, hypnotics, such as benzodiazepines, barbiturates, paraldehyde, glutethimide, chloral hydrate and alcohols.

Anti-convulsants such as barbiturates, hydantoins, oxazolindiones, succinimides, benzodiazepines.

Antipsychotics, such as phenothiazines, fluorobutyrophenones, β -aminoketones, CNS depressants with skeletal muscle relaxant properties.

4. Central Nervous System Stimulants

Analeptics, picrotoxin, methylxanthines, monoamine oxidase inhibitors, tricyclic compounds. Indolethylamines, 2-phenylethylamines.

5. Diuretics

Carbonic anhydrase inhibitors, thiazide and thiazide like diuretics, high ceiling or loop diuretics, potassium sparing diuretics and miscellaneous compounds such as mercaptopurine, meralluride, Thiazides, spironolactone, theophylline, furosemide, acetazolamide, ethacrynic acid, triamterene.

6. Antiviral agents such as acyclovir, tromantadine hydrochloride, ribavirin**7. Immunosuppressive agents** such as azathioprine, cyclosporine**8. Anti-Neoplastic Agents**

Alkylating agents, antimetabolites, antibiotics, plant products, miscellaneous compounds, hormones, immunotherapy, such as methotrexate, 5-fluorouracil, actinomycines, anthracyclines, vincristine, tamoxifen.

9. Anti-Histamines

H₁-antagonists, H₂-antagonists, aminoalkyl ethers, ethylenediamines, propylamine derivatives, phenothiazine derivatives, piperazine derivatives, such as diphenhydramine, pyrilamine, promethazine, cyclizine, terfenadine, sucralfate, cimetidine, ranitidine, omeprazole.

Books Recommended

1. Principles of Medicinal Chemistry, William O. Foye, Thomas L. Lemke and David A. Williams, Wolters Kluwer, Lippincott Williams & Wilkins 7th edition, (2013).
2. Textbook of Organic Medicinal and Pharmaceutical Chemistry, Wilson and Gisvold's.. 12th edition, Wolters Kluwer, Lippincott Williams & Wilkins (2011).
3. Introduction to Medicinal Chemistry: How Drugs Act and Why, Alex. Gringuaz, John Wiley & Sons Inc. (1997).
4. Burger's Medicinal Chemistry and Drug Discovery edited by E. Wolf, 6th edition, Wiley-Inter Science, New York (2003).
5. The Organic Chemistry of Drug Design and Drug Action, Richard B. Silver. Academic Press, Pnen, Inc., USA (1992).

SECOND SEMESTER

First Professional

PHC-304	Organic and Inorganic Chemistry (Practical)	Cr. Hrs. 3
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1. Identification of unknown simple organic compounds containing acidic, basic and neutral functional groups.
2. Estimation of functional groups in the following drugs.
Sulpa drugs, aspirin, paracetamol, benzyl penicillin, bromisoval, methenamine.
3. Determination of iodine and saponification value, and ash content.
4. Preparation of some organic and inorganic compounds
Acetanilide, iodoform, nitrophenol, 3-nitrophthalic acid, benhydrol, 2, 4-dinitro-chlorobenzene, ferrous sulfate, aluminum hydroxide, magnesium carbonate, sodium carbonate.

Books Recommended

1. A Guide Book to Mechanisms in Organic Chemistry Peter Sykes.. Fourth edition, Longman Group Ltd. (1978).
2. Pharmaceutical Analysis – A Textbook of Pharmacy, Student and Pharmaceutical Chemistry Watson D.G., Churchill Livingston (1999).
3. Introduction to Organic Chemistry By William H. Brown Saunders College Publishing 3rd edition (2002).

4. Organic Chemistry Graham Solomons T.W. Fifth edition, John Wiley & Sons, Inc. (1992).
5. Practical Pharmaceutical Chemistry Backet and Stenlake, Royal Press, London (1998).

PHC-306

Pharmaceutical Statistics

Cr. Hrs. 3

1. What is statistics? Importance's of statistics. What is biostatistics? Application of statistics in biological and pharmaceutical sciences. How samples are selected.
2. Organizing and displaying data: Variables, quantitative and qualitative variables, univariate data, bivariate data. Frequency table, diagrams, pictograms, simple bar charts, multiple bar charts, histograms.
3. Summarizing data and variation: The mean, the median, the mode, the variance and standard deviation, coefficient of variation and skewness.
4. Probability: Definitions of probability, rules of probability. Distributions (binomial poisson and normal probability distributions)
5. Test of hypothesis and significance: Statistical hypothesis, level of significance, test of significance. Confidence intervals. Test involving binomial and normal distributions. Student "t" and Chi-square distribution test of significance based on "t" and chi square distributions
6. Curve Fitting: Fitting a straight line, fitting of parabolic or high degree curve.
7. Regression and correlation: simple linear regression model. Correlation co-efficient
8. Analysis of variance: One-way classification, partitioning of suen of squares and degrees of freedom. Multiple compression tests such as LSD. The analysis of variance models

Books Recommended

1. Beth Dawson-Saunder. Basic and Clinical Biostatistics, 3rd Edition, A Lange Medical Book. London.
2. Biostatistics, A Foundation Analysis in the Health Sciences, Danniell, W.W., 4th edition Bernard Rosner (1987).
3. Fundamentals of Biostatistics, 4th Edition Wadsworth Publishing Company, USA (1995).
4. Introduction to Statistics, Ronalde Walpole, 3rd edition (1990).
5. Remington Practical of the Science and Pharmacy, Mack Publishing Company, Eston, Pennsylvania, USA.

Second Professional

PHC-406

Physical Chemistry – I (Practical)

Cr. Hrs. 2

1. Determination of specific gravity of liquids/syrups using an Ostwald pycnometer.
2. Determination of the viscosity of a liquid by Ostwald viscometer.
3. Determination of the percent composition of a mixture of ethanol and water by viscometric method.
4. Determination of the parachor value of an organic liquid.
5. Determination of the surface tension of a pure liquid by the drop count method.

6. Determination of the percentage composition of mixture of ethanol and water by surface tension method.
7. Determination of interfacial tension between benzene and water by the drop count method.
8. Determination of solubility of benzoic acid over a range of temperatures and calculation of its heat of solution.
9. Determination of the mutual solubility curve of phenol and water.
10. Preparation of buffer solutions and measurement of pH.
11. Determination of variation of miscibility with temperature.
12. Determination of the partition coefficient of I_2 between CCl_4 and H_2O .

Books Recommended

1. Pharmaceutical Analysis, Higuchi T. and Brochmann Hanssen E., Interscience, New York.
2. Physical Chemistry Barrow G.M., McGraw-Hill, London (1973).
3. Physical chemistry by Fazal I Hussain, 3rd Edition (1991).
4. Physical Chemistry, Barrow G.M., McGraw-Hill, London (1973).
5. Text Book of Physical Chemistry by Sameul Glasstone 2nd Edition MC Millan & Company Ltd. (1946).

PHC-408

Physical Chemistry – II

Cr. Hrs. 3

1. **Electrochemistry**
Definition, resistance, conductance, specific conductance, conductivity and its measurement, conductometric titrations, electrochemical cells, determination of pH and redox potentials, electrophoresis and electrodialysis.
2. **Photochemistry**
Definition light absorption and excitation of organic compounds, photophysical and photochemical processes, photochemical reactions, photosensitisation and photocatalysis, photolysis of medicinal compounds and photostabilization, high energy radiations.
3. **Solid and Crystalline State**
Formation of solids, types of solids, nature of amorphous and crystalline solids, crystal systems, determination of crystal structure, polymorphism.
4. **Chemical Kinetics**
Rates and order of reactions, influence of temperature and other factors on reaction rates, acid-base catalysis, decomposition of medicinal compounds, accelerated stability analysis, kinetics of enzyme catalysed reactions.
5. **Thermodynamics**
First and second law of thermodynamics, thermochemistry, relationship between free energy, entropy and equilibrium constant, free energy functions and applications.

Books Recommended

1. Physical Chemistry P.W. Atkins 5th edition W.H. Freeman and company New York 2000
2. Physical Pharmacy Martin A.N., 4th Edition, Lea and Febiger, Philadelphia (1993).
3. Practical Pharmaceutical Chemistry Backet and Stenlake., Royal Press, London (1998).

4. Remington Practical of the Science and Pharmacy, Mack Publishing Company, Easton, Pennsylvania, USA.

Third Professional

PHC-506 Preparation and Quality Control (Practical) Cr. Hrs. 3

1. Preparation/synthesis, identification and purity determination of the following important pharmaceutical compounds as mentioned in B. P and U. S. P.
Paracetamol, salicylic acid, methyl salicylate, azobenzene, benzoic acid. 5-Hydroxy-1, 3-benzoxazol-2-one, aspirin, p-nitrosophenol, 3-nitrophthalic acid, o-Chlorobenzoic acid.
2. Limit tests as mentioned in pharmacopoeias for lead, arsenic, chlorides, sulphates, iron and heavy metals

Books Recommended

1. Laboratory Techniques and Organic Preparations, Nousheen Mushtaq & Shamim Akhtar, Manual for PHC 506.
2. A Guide Book to Mechanisms in Organic Chemistry Peter Sykes. Fourth edition, Longman Group Ltd. (1978).
3. Analytical Application of Spectroscopy Greaser C.S. and Davies A.M.C. Royal Society of Chemistry, London (1988).
4. Introduction to organic chemistry By William H. Brown Saunders College Publishing 3rd edition (2002).
5. Remington Practical of the Science and Pharmacy, Mack Publishing Company, Easton, Pennsylvania, USA.

PHC-508 Pharmaceutical Analysis - I Cr. Hrs. 3

To study the principles and applications of the following methods in the drug analysis and development

1. **Chemical Methods**
Titrimetric methods, Gravimetric methods, Solvent extractions methods.
2. **Electro Chemical Methods**
Potentiometry, Polarography, Conductometry.
3. **Optical Methods**
Refractometry, Polarimetry, Fluorimetry.
4. **Radiochemical Methods**
Introduction to Radiopharmaceuticals, Analysis, Quality Control, Stability, Applications.

Books Recommended

1. Instrumental Methods of Chemical Analysis by Golden W. Evings, McGraw Hill London 5th Edition (1985).
2. Pharmaceutical Analysis – A Textbook of Pharmacy, Student and Pharmaceutical Chemistry Watson D.G. Churchill Livingston (1999).
3. Principle of Instrumental Analysis, Skoog Holter and Nieman 5th Edition (2000).
4. Quantitative Analysis, Theory and Practice by Lawrence W. Pott Harper and Raw Publisher New York (1987).
5. Vogel's Text Book of Quantitative Organic Chemistry, Longman, London. 6th Edition (2006).

Fourth Professional

PHC-608

Pharmaceutical Analysis – II

Cr. Hrs. 3

To study the principles and applications of the following methods in the drug analysis and development

1. **Spectroscopic Methods**
Spectrophotometry (UV, Visible and Infrared), atomic absorption spectroscopy, mass spectrometry, nuclear magnetic resonance spectrometry, x-ray spectroscopy.
2. **Chromatographic Methods**
Thin layer chromatography, column chromatography, gas-liquid chromatography, high performance liquid chromatography, ion-exchange chromatography, size exclusion or gel chromatography.

Books Recommended

1. Principle of Instrumental Analysis, Skoog Holter and Nieman 5th Edition (2000).
2. Chromatographic Methods, R-Stock and CBE Rice 2nd Edition, Chapman and Hall Ltd and Science (1966).
3. Instrumental Methods of Chemical Analysis by Golden W. Evings, McGraw Hill London 5th Edition (1985).
4. Modern Infra Red Spectroscopy, Barbara Stuart Publisher ACOL John Wily and Sons (1996).
5. Vogel's Text Book of Quantitative Organic Chemistry, Longman, London. 6th Edition (2006).

PHC-610

Medicinal Chemistry- I

Cr. Hrs. 3

1. **Introduction to medicinal chemistry.**
2. **Classification of drugs** on the basis of sources, structure, site of action and mode of action.

3. **Preparation and properties of medicinally important heterocyclic compounds** such as: pyrrol, furan, thiophene, pyridine, pyrimidine and pyrazine. Five and six membered ring containing one heteroatom: indole, quinoline and Isoquinoline.
4. **Drug Discovery and designing:**
 - a. Introduction of lead : qualities and requirements
 - b. Finding “lead” compounds
5. **Drug Development:** Lead Modification
 - a. Pharmacophore and modifications in structure
 - b. Physicochemical parameters affecting to drug development
6. **Metabolism of Drug**
 - a. Introduction
 - b. Reactions involved in Metabolism
 - i. Phase-I transformation
 - ii. Phase-II transformation
7. **Structural features of Drugs**
 - a. Stereochemistry and drug action
 - b. Configuration and drug receptor interaction
 - c. Conformation and drug receptor interaction
8. **Forces involved in drug receptor interaction**
9. **Introduction and types of drug targets**
10. **Drug Receptor interaction; Theories**
11. **Structure activity relationship**
12. **Alkaloids** – atropine, morphine and related compounds (codeine, thebaine), ergotamine, reserpine, ephedrine.
13. **Hormones** (steroidal and proteinous): Testosterone, progesterone, estrogen, aldosterone, cortisol, insulin, glucagon, oxytocin and vasopressin.
14. **Modern concept of rational drug design**
 - a. Computational Chemistry
 - b. Combinatorial Chemistry
 - c. Biotechnology and drug discovery

Books Recommended

1. An introduction to medicinal chemistry by Graham L. Patrick Oxford University Press 2nd edition (2000).
2. Principles of Medicinal Chemistry, William O. Foye, Thomas L. Lemke and David A. Williams, Wolters Kluwer, Lippincott Williams & Wilkins 7th edition, (2013).

3. Textbook of Organic Medicinal and Pharmaceutical Chemistry, Wilson and Gisvold's.. 12th edition, Wolters Kluwer, Lippincott Williams & Wilkins (2011).
4. Introduction to Medicinal Chemistry: How Drugs Act and Why, Alex. Gringuaz, John Wiley & Sons Inc. (1997).
5. Introduction to organic chemistry By William H. Brown Saunders College Publishing 3rd edition
6. The Organic Chemistry of Drug Design and Drug Action, Richard B. Silver. Academic Press, Pnen, Inc., USA (1992).

Fifth Professional

PHC-710

Medicinal Chemistry –III

Cr. Hrs. 3

1. **To study the chemistry, structure, mechanism of action, structure activity relationship and therapeutic applications of the following**
 - a. **Antibacterial / Antibiotics**
Sulphonamides such as sulfamethoxalor, sulfadiazine, sulfafurazole. Penicillines, Cephalosporine, Strepto-mycin, Chloramphenicol, Tertacyclines, Kamamycin and Erythromycin.
 - b. **Antimalarial agents** such as 4-aminoquinolines, 8-aminoquinolines, 9-amino acridine, biguanides, pyrimidine, analogues, mefloquine, cinchona alkaloids.
 - c. **Anthelmintics** such as piperazine derivatives, thiabendazole, mebendazole, pyrantal.
 - d. **Antitubercular drugs** such as ethambutol, isonicotinic acid, hydrazid rifampicin, pyrizinamide, cycloserine, thambutol, ethionamide, streptomycin, kanamycin, para aminobanzoic acid, rifamycin, fluoroquinolones
 - e. **Hypoglycemic agents** such as sulfonylureas tolbutamide, chlorpropamide, acetohexamide, glipizide, glyburide
 - f. **Cardiovascular agents:** Antianginal agents and vasodilators, antiarrhythmic drugs, antihypertensive agents, angiotensin-converting enzymes inhibitors, antihyperlipidemic agents, anticoagulants
 - g. **Adrenergic and cholinergic agents neurotransmitters, receptors, agonists and antagonists**
 - h. **Vitamins** (water and fat soluble). B₁, B₂, B₆, B₁₂, folic acid, nicotinic acid, biotin, pantothenic acid, ascorbic acid A, D, E and K
2. **To study the Occurrence, properties, preparation and applications of the following medicinally active inorganic compounds.**

Aluminium Hydroxide, Ammonium Chloride, Sodium Carbonate, Magnesium Carbonate, Lithium Carbonate, Sodium Nitrite , Calcium Gluconate, Ferrous Fumarate, Ferrous Sulfate and Silver nitrate

Books Recommended

1. Principles of Medicinal Chemistry, William O. Foye, Thomas L. Lemke and David A. William, Wolters Kluwer, Lippincott Williams &Wilkins 7th edition, (2013).
2. Textbook of Organic Medicinal and Pharmaceutical Chemistry, Wilson and Gisvold's.. 12th edition, Wolters Kluwer, Lippincott Williams &Wilkins (2011).
3. Introduction to Medicinal Chemistry: How Drugs Act and Why, Alex. Gringuaz, John Wiley & Sons Inc. (1997).
4. Burger's Medicinal Chemistry and Drug Discovery edited by E. Wolf, 6th edition, Wiley-Inter Science, New York (2003).
5. The Organic Chemistry of Drug Design and Drug Action, Richard B. Silver. Academic Press, Pnen, Inc., USA (1992).

Department of Pharmacology

Pharm-D Courses

First Year					
1st Semester			2nd Semester		
Course No.	Title of Course	Cr. Hr.	Course No.	Title of Course	Cr. Hr.
PHL-305	Islamic Ideology- Pakistan studies	2	PHL-308	Pharmacology Physiology & Histology- II	3
PHL-307	Pharmacology - Physiology & Histology-I	3	PHL-310	Pharmacology - Pharmaceutical Biochemistry II	3
PHL-309	Pharmacology - Pharmaceutical Biochemistry-I	3	PHL-314	Pharmacology - Anatomy	2
PHL-311	Pharmacology - Pharmaceutical Biochemistry (Lab)	3	-	-	-
Second Year					
PHL-407	Pharmacology - Physiology & Histology (Lab)	3	PHL-410	Pharmacology - Systemic Pharmacology – I	3
PHL-409	Pharmacology & Therapeutics	3	-	-	-
PHL-413	Pharmacology - Pathology	2	-	-	-
Third Year					
PHL-507	Pharmacology - Systemic Pharmacology –II	3	PHL-510	Pharmacology - Pharmacology Lab - I	3
PHL-509	Pharmacology - Pathology (Lab)	2	-	-	-
Fourth Year					
PHL-609	Pharmacology - Systemic Pharmacology - III	3	PHL-612	Pharmacology Lab - II	3
Fifth Year					
PHL-711	Pharmacology - Clinical Pharmacology	2	PHL-712	Pharmacology - Toxicology	2
Total 18 Courses making 48 Credit Hours in five years.					

FIRST SEMESTER

First Professional

PHL-305 **Islamic Ideology Pakistan Studies** **Cr. Hrs. 2**

1. The Need Religion

A critical analysis of the sources of human knowledge and importance of why (Divine revelation).

2. Islamic Concept of Life

Islamic concept of universe, the position of mankind, the earth, the goal for men's activities.

3. Islamic Beliefs

Islamic and aqida the role of iman in character building and in the development civilization. Aqida-tawhid (beliefs in unity of God) its details and its impact on character. Iman bil Malaikah (belief in angels). Aqida risalat (belief in prophethood) its details. Its importance In the development of Islamic civilization, the distinguishing features of Mohammad (PBUH) Prophethood, the doctrine of the last prophet. Iman bil kutub (Belief in the revealed books). Aqida Akhlat (Belief in the life hereafter) its details, quranic style of arguments on the life hereafter. The impact of Aqida Akhlat on individual and society. A comparison and Islamic concept of Ibadat with other religions. Salat (Prayer). Zakat, its philosophy, rates and minimum zakat amount, its impact on economy. Sawm (Fasting). Haj (Pilgrimage): Manasik and its importance.

4. Pakistan Studies

Ideology of Pakistan and its basic elements, two nation theory; Aims and objective for Establishment of Pakistan; The Khilafat Movement. The struggle for Islamic System (with the special reference to constitution of Pakistan 1973); The common problems of Muslim Ummah and their solutions.

5. Islamic Moral Values

Philosophy of morality in Islamic moral Values.

Books Recommended

1. Kursheed Ahmed, Islam ka Nazariya hayat
2. Shiblee Nuamani/Syed Suleman Nadvi. Seerat-un-Nabi (SAWW) Jild-e-Kamil

PHL-307 **Physiology & Histology-I** **Cr. Hrs. 3**

1. Physiology of Nerve and Muscle

Chemical changes during muscle contraction, nerve action potential, skeletal, smooth muscle and cardiac excitation contraction.

2. Blood

Composition, functions and genesis of formed elements. RBC, WBC and platelet. Fate of RBC, jaundice, reaction of blood, blood groups. Rh factors, ESR, blood volume, function of spleen, blood coagulation, hemophilia, classification of anemias.

3. Circulatory System

Properties of cardiac muscles, origin and conduction of heart beat, cardiac cycle, ECG, heart sounds, cardiac output, stroke volume and heart rate. Nerve supply to heart, coronary, pulmonary and skin circulation. Blood pressure, vasomotor center. Arterial pulse, venous pulse, hemorrhage, circulatory changes in exercise, composition and circulation of lymph, shock.

4. Skin

Structure and functions of skin, temperature regulation.

5. Digestive System

Mastication, deglutition, digestive juices (gastric, pancreatic, bile and intestinal juices) their composition, function and mechanism of secretions. Movement of stomach and intestine, function of large intestine, defecation, functions of liver and gall bladder.

Books Recommended

1. Arthur C. Guyton and John E. Hall. Textbook of Medical Physiology. 13th Edition, 2015, W.B. Saunders, Philadelphia.
2. Barbara Young philipwoord ford, Geraldine o' Dowd Wheater's Functional Histology: A Text and Colour Atlas. 6th Edition, 2013.
3. Douglas F. Paulsen. Basic Histology: Examination and Board Review. 5th edition, 2010, Prentice Hall Internal Inc.
4. Frederic H. Martini. Fundamentals of Anatomy and Physiology. 10th Edition, 2014, Prentice Hall, New Jersey.
5. Gerard J. Tortora & Bryan Derrickson. Principles of Anatomy and Physiology. 14th Edition, 2013, John Wiley & Sons, New York.

PHL-309

Pharmaceutical Biochemistry-I

Cr. Hrs. 3

1. Introduction and Basic Biochemical Principles

Role of pharmaceutical biochemistry in the health profession, nature of biochemical reactions.

2. Basic Chemistry of Biomolecules

Carbohydrates; Chemistry, classification, reactions, optical activity, biological and pharmaceutical importance of carbohydrates.

Lipids; Chemistry of fatty acids and lipids classifications, (saponifiable and non saponifiable lipids, simple, complex and derived). Reactions of fatty acids and other lipids. Essential fatty acids, biological and pharmaceutical importance of lipids.

Proteins and Amino acids; Chemistry, classification, reactions of proteins and amino acids. Organizational level, macromolecular nature, biological and pharmaceutical importance of proteins and amino acids.

3. Metabolic Fate of Biomolecules

Carbohydrates; Brief digestion and absorption, aerobic and anaerobic breakdown of glucose, glycolysis, pentose phosphate pathway, glycogenolysis, gluconeogenesis, citric acid cycle, energetics of various metabolic processes.

Lipids; Brief digestion and absorption, oxidation of fatty acids through beta oxidation, Biosynthesis of fatty acids, neutral lipids, and cholesterol.

Proteins and Amino-acids; Brief digestion and absorption, metabolism of essential and non essential amino acids, Biosynthesis and catabolism of Haemins and porphyrin compounds.

4. Bioenergetics

Principles of bioenergetics, electron transport chain and oxidative phosphorylation.

5. Enzymes

Chemistry, classification, mode of action, kinetics (Michaelis Menten Equation and some modifications), Inhibition, activation, specificity, allosteric enzymes. Factors affecting the rate of an enzyme catalyzed reaction, Biological and pharmaceutical importance, mechanism of action of some important enzyme (Chymotrypsin, Ribonuclease).

Books Recommended

1. Alisa Peet, Michael A. Lieberman, Allan Marks. Marks' Basic Medical Biochemistry. 4th edition, 2012, Lippincott Williams & Wilkins.
2. David L. Nelson , Michael M. Cox. Lehninger Principles of Biochemistry. 6th Edition, 2012, W.H. Freeman.
3. MN Chatterjee & Rana Shinde. Text book of medical biochemistry. 8th Edition. 2011 Jaypee publishers.
4. Pamela C. Champe & Richard A. Harvey. Lippincott's Illustrated Review: Biochemistry. 6th Edition, 2013, Lippincott Williams & Wilkins, Philadelphia.
5. Robert K. Murray. Harper's Illustrated Biochemistry. 30th Edition, 2015, Lange Medical Books, McGraw Hill, New York.

PHL-311

Pharmaceutical Biochemistry (Lab)

Cr. Hrs. 3

1. Qualitative Analysis

Carbohydrates, amino acids, peptides and proteins. Lipids and sterols (Cholesterol) bile salts and bilirubin. Blood sugar analysis, uric acid, bilirubin, cholesterol and creatinine.

2. Quantitative Analysis

Carbohydrates–Glucose (reducing sugar) and any other carbohydrate using Benedict and anthrone method. Amino acids. Peptides and proteins using Biuret and Ninhydrin (Spectrophotometric) method. Analysis of normal and abnormal constituents of urine sugar. Uric acid, bilirubin, cholesterol and creatinine

Books Recommended

1. Alisa Peet, Michael A. Lieberman, Allan Marks. Marks' Basic Medical Biochemistry. 4th edition, 2012, Lippincott Williams & Wilkins.
2. David L. Nelson, Michael M. Cox. Lehninger Principles of Biochemistry. 6th Edition, 2012, W.H. Freeman.
3. MN Chatterjee & Rana Shinde. Text book of medical biochemistry. 8th Edition. 2011 Jaypee publishers.
4. Pamela C. Champe & Richard A. Harvey. Lippincott's Illustrated Review: Biochemistry. 6th Edition, 2013, Lippincott Williams & Wilkins, Philadelphia.
5. Robert K. Murray. Harper's Illustrated Biochemistry. 30th Edition, 2015, Lange Medical Books, McGraw Hill, New York.

Second Professional

PHL-407 Physiology & Histology (Lab) Cr. Hrs. 3

1. **Introduction to Experimental Physiology**
2. **Blood**
Determination of hemoglobin. Determination of ESR, RBC count, WBC count, differential leucocytes count, bleeding and clotting time, blood groups.
3. **Respiration**
Estimation of vital capacity and its relation to posture and standard vital capacity. Determination of tidal volume. Demonstration of artificial respiration.
4. **C.V.S.**
Recording of arterial pulse, recording of arterial B.P.
5. **Eye**
Visual acuity, far vision, near vision and field of vision.
6. **C.N.S.**
Nerve muscle preparation in frog, effect of temperature on muscle, Demonstration of special reflexes.
7. **Histology**
Demonstration, preparation and staining of the slides, histological examination of slides, epithelium, connective tissue, muscle tissue, organ system - lungs, kidney, appendix, skin, gall bladder, stomach, intestine.

Books Recommended

1. Arthur C. Guyton and John E. Hall. Textbook of Medical Physiology. 13th Edition, 2015, W.B. Saunders, Philadelphia.
2. Barbara Young, Philip Woodford, Geraldine O'Dowd Histology: A Text and Colour Atlas 6th Edition, 2013.
3. Douglas F. Paulsen. Basic Histology: Examination and Board Review. 5th Edition, 2010, Prentice Hall International Inc.

4. Frederic H. Martini. Fundamentals of Anatomy and Physiology. 10th Edition, 2014, Prentice Hall, New Jersey.
5. Gerard J. Tortora & Bryan Derrickson. Principles of Anatomy and Physiology. 14th Edition, 2013, John Wiley & Sons, New York.

PHL-409

Pharmacology & Therapeutics

Cr. Hrs. 3

1. Introduction

History and scope of Pharmacology, classification of Pharmacology, classification of drugs and their sources.

2. Definitions

Bioavailability, bioequivalence, therapeutic index, potency, efficacy, risk benefit ratio, selective toxicity, plasma half-life, dose response curve, desensitization and tachyphylaxis.

3. Drugs Delivery System

Advantages and disadvantages of oral medication. Advantages and disadvantages non-oral medication.

4. Pharmacokinetics

Drug solubility and passage of drugs across body membranes, plasma concentration of drugs and various factors affecting it. Factors affecting absorption, distribution, biotransformation and excretion.

5. Pharmacodynamics

Drug receptors and theories, agonist and antagonists, mechanism of drug action, specificity of drug action, and factors modifying the action.

6. Drugs Acting on Blood

Antianemic (Vit B₁₂, folic acid, iron). Coagulants and anticoagulants.

7. Autonomic Nervous System

Introduction to autonomic pharmacology.

8. Drugs Acting on ANS

Sympathetic agonists, sympathetic antagonists, parasympathetic agonists and antagonists, anticholinesterases, ganglion blockers and neuromuscular blockers.

Books Recommended

1. [Anthony Trevor](#), [Bertram Katzung](#), [Susan Masters](#), [Marieke Knuidering-Hall](#). Katzung & Trevor's Pharmacology Examination and Board Review, 11th Edition, 2015. Lange Medical Books.
2. Bertram G. Katzung, [Susan Masters](#), [Anthony Trevor](#). Basic and Clinical Pharmacology, 13th Edition, 2014. A Lange Medical Book. London.
3. David E. Golan, Armen H. Tashjian, Jr. Ehrin J. Armstrong, April W. Armstrong. Principles of Pharmacology "The Pathophysiologic Basis of Drug Therapy". 3rd Edition (2011). Lippincott Williams & Wilkins.
4. Goodman & Gilman's. The Pharmacological Basis of Therapeutics 12th Edition, 2010. McGraw-Hill, USA.

5. Rang H.P., Dale M.M. Rang & Dale's Pharmacology, 8th Edition, 2015. Churchill Living Stone, England.

PHL-413 Pharmacology-Pathology Cr. Hrs. 2

- 1. Scope of Pathology and Concept of Diseases**
- 2. Definitions and Terminologies**
- 3. Response of Body to Injury and Infection**
Acute inflammation, chronic inflammation. Immunity, allergy and hyper sensitivity.
- 4. Specific Diseases**
Peptic and duodenal ulcer, hypertension. M.I., SLE, Nephrotic syndrome, COPD.
- 5. Diagnosis of Cancer**
Fate, survival and prognosis of tumors. Leukemia, malignant carcinoma, sarcoma lymphoma.

Books Recommended

1. David E. Golan, Armen H. Tashjian, Jr. Ehrin J. Armstrong, April W. Armstrong. Principles of Pharmacology. "The Pathophysiologic Basis of Drug Therapy". 3rd Edition (2011).
2. Martin Gwent Lewis and Thomas K. Barton. Appleton & Lange's Review of General Pathology. 4th Edition, 2002, Prentice Hall International Inc.
3. Michael D. Randall and Karen E Neil. Disease Management, 2nd Edition, 2008.
4. Ramzi S. Cotran, Vinay Kumar & Stanley L. Robbins. Robbins & Cotran Pathologic Basis of Disease. 9th Edition, 2014. W.B. Saunders Company, Philadelphia.
5. Russell J. Greene and Norman D. Harris. Pathology and Therapeutics for Pharmacists: A Basis for Clinical Pharmacy Practice. 3rd edition, 2008. Chapman & Hall, London

Third Professional

PHL – 507 Systemic Pharmacology-II Cr. Hrs. 3

- 1. Drugs Acting on Cardio Vascular System**
Angina and antianginal drugs. Thrombosis and thrombolytic agents, Congestive heart failure and its treatment. Cardioactive glycosides, Bipyridines, β -adrenergic blockers. Xanthine derivatives, vasodilators. Antiarrhythmic drugs, Hyperlipidemia and hypocholesterolemic agents. Hypertension, antihypertensives and diuretics.
- 2. Dermatological Agents**
Factors affecting topical absorption of drugs, Pharmacology of dermatological agents.
- 3. Drugs Acting on Respiratory System**
Drugs used for Cough: Antitussives, demulcents, steam inhalation, local anesthetics. Narcotic antitussives, non-narcotic antitussives, expectorants and mucolytic agents.
Bronchodilators, Corticosteroids and other anti-inflammatory drugs, Muscarinic, receptor antagonists, Mast cell stabilizers; β agonists, Leukotriene inhibitors.

4. **Anti Neoplastics** Drug used for treatment of cancer including Alkylating agents, Mitotic Spindle poisons, Podophyllotoxins, Anti Biotic, Miscellaneous, Topoisomerase inhibitors and Mono Clonal Antibodies.

Books Recommended

1. [Anthony Trevor](#) , [Bertram Katzung](#) , [Susan Masters](#), [Marieke Knudering-Hall](#). Katzung & Trevor's Pharmacology Examination and Board Review, 11th Edition, 2015. Lange Medical Books.
2. Bertram G. Katzung, [Susan Masters](#), [Anthony Trevor](#). Basic and Clinical Pharmacology, 13th Edition, 2014. A Lange Medical Book. London.
3. David E. Golan, Armen H. Tashjian, Jr. Ehrin J. Armstrong, April W. Armstrong. Principles of Pharmacology "The Pathophysiologic Basis of Drug Therapy". 3rd Edition (2011). Lippincott Williams & Wilkins.
4. Goodman & Gilman's. The Pharmacological Basis of Therapeutics 12th Edition, 2010. McGraw-Hill, USA.
5. Rang H.P., Dale M.M. Rang & Dale's Pharmacology, 8th Edition, 2015. Churchill Living Stone, England.

PHL-509

Pharmacology Pathology (Lab)

Cr. Hrs. 2

1. **Study of Pathological Slides of various Pathological Conditions**
Acute inflammation; chronic inflammation; chronic specific inflammation. Different types of degeneration. Thrombosis, embolism, infarction, necrosis, gangrene, hyperplasia, metaplasia, pigmentation, calcification, C.B.C., papilloma, adenoma, chondroma, fibroma, neofibroma, sq. cell carcinoma, Basal cell carcinoma, transitional cell carcinoma.
Adenocarcinoma, fibrocarcinoma, rhabdomyosarcoma, leiomyosarcoma, lymphosarcoma, liposarcoma, reticular cell sarcoma, Hodgkin's disease, breast carcinoma, osteogenic, Sarcoma, osteoclastoma.
2. **Examination of Different Body Fluids in various Pathological Conditions**
Urine complete examination, stool examination, blood complete examination. Semen examination, cerebrospinal fluid examination, pericardial fluid examination, pleural fluid examination, aseptic fluid examination, blood sugar, blood urea, blood cholesterol etc.
3. **Tests for various Specimens of Clinical Importance**
Techniques of clinical blood examination for various diseases, gastric analysis, Tests for liver function and renal function. Test for endocrine abnormalities, biopsies and cytological techniques.

Books Recommended

1. David E. Golan, Armen H. Tashjian, Jr. Ehrin J. Armstrong, April W. Armstrong. Principles of Pharmacology. "The Pathophysiologic Basis of Drug Therapy". 3rd Edition (2011).
2. Martin Gwent Lewis and Thomas K. Barton. Appleton & Lange's Review of General Pathology. 4th Edition, 2002, Prentice Hall International Inc.
3. Ramzi S. Cotran, Vinay Kumar & Stanley L. Robbins. Robbins & Cotran Pathologic Basis of Disease. 9th Edition, 2014. W.B. Saunders Company, Philadelphia.

4. Russell J. Greene and Norman D. Harris. Pathology and Therapeutics for Pharmacists: A Basis for Clinical Pharmacy Practice. 3rd edition, 2008. Chapman & Hall, London
5. [Stephen J. McPhee](#), [Maxine Papadakis](#), [Lawrence M. Tierney](#) . Current Medical Diagnosis and Treatment, 54th Edition, 2015, Lange Medical Books, McGraw-Hill Medical Publishing Division, London.

Fourth Professional

PHL-609

Systemic Pharmacology - III

Cr. Hrs. 3

1. **Drug Acting on Central Nervous System**
Hypnotic and sedatives, analgesics, narcotic analgesics and opioids antagonists, anxiolytics, antipsychotics, antidepressants, antimanic, cerebral stimulants, spinal cord stimulants, drug treatment of epilepsy, drug treatment of parkinsonism and other movement disorders.
General and local anesthetics
2. **Insulin, thyroxin and other agents affecting endocrine function**
3. **Ocular Pharmacology**
Use of autonomic agents in eyes, chemotherapy of diseases in eye.
Use of immunomodulatory drugs for ophthalmic therapy; Use of anesthetics in ophthalmic procedures.
4. **Oxytocic drugs**
5. **Drugs Acting on Reproductive System**
Contraceptives; Fertility drugs; Testosterone and contraception in males; erectile dysfunction and pharmacotherapy.

Books Recommended

1. [Anthony Trevor](#) , [Bertram Katzung](#), [Susan Masters](#), [Marieke Knudering-Hall](#). Katzung & Trevor's Pharmacology Examination and Board Review, 11th Edition, 2015. Lange Medical Books.
2. Bertram G. Katzung, [Susan Masters](#), [Anthony Trevor](#). Basic and Clinical Pharmacology, 13th Edition, 2014. A Lange Medical Book. London.
3. David E. Golan, Armen H. Tashjian, Jr. Ehrin J. Armstrong, April W. Armstrong. Principles of Pharmacology "The Pathophysiologic Basis of Drug Therapy". 3rd Edition, 2011. Lippincott Williams & Wilkins.
4. Goodman & Gilman's. The Pharmacological Basis of Therapeutics 12th Edition, 2010. McGraw-Hill, USA.
5. Rang H.P., Dale M.M. Rang & Dale's Pharmacology, 8th Edition, 2015. Churchill Living Stone, England.

Fifth Professional

PHL-711

Clinical Pharmacology

Cr. Hrs. 2

- 1. Introduction to Clinical Pharmacology**
Terminology, basic components and scope.
- 2. Role of Drug Monitoring in Therapeutics**
Patient profile, diseases profile, drug profile, monitoring responses, monitoring plasma concentration.
- 3. Factors Affecting Drug Response**
Pharmacogenetics, drug interactions.
- 4. Development of New Drugs**
Process of drug development, preclinical studies, types of clinical trials, choice of patients, exclusion criteria of patients.
- 5. Drugs in Pregnancy**
Prescribing in pregnancy, harmful effects on fetus, pharmacokinetics in pregnancy.
- 6. Drugs in Infants and Children**
Practical aspects of prescribing drugs, pharmacokinetics.
- 7. Drugs in Elderly**
Pharmacokinetics changes, Pharmacodynamic changes.
- 8. Drug Toxicity**
Adverse drug reactions, monitoring adverse drug reactions, risk benefit ratio.

Books Recommended

1. [Arthur J. Atkinson Jr](#), [Shiew-Mei Huang](#), Juan J.L. Lertora, [Sanford P. Markey](#). Principles of Clinical Pharmacology. 3rd edition, 2012, Academic Press.
2. P.V. Rataboli. Clinical Pharmacology and Rational Therapeutics, 2nd edition, 2010, Ane Books Pvt Ltd.
3. [Gerard A. McKay](#), [Matthew R. Walters](#). Clinical Pharmacology and Therapeutics .9th edition, 2013, Wiley Blackwell.
4. Mary Anne Koda Kimble. Koda-kimble & Young's Applied Therapeutics: "The Clinical Use of Drugs" 10th Edition, 2012. Lippincott Williams & Wilkins USA.
5. Roger Walker and Cate Whittlesea. Clinical Pharmacy and Therapeutics, 5th Edition (2011).

SECOND SEMESTER

First Professional

PHL-308

Physiology and Histology-II

Cr. Hrs. 3

1. Respiratory System

Mechanics of respiration, intrathoracic, intrapulmonary pressure. Pulmonary ventilation. Lungs volume and capacities. Composition of inspired air, expired air and alveolar air, carriage of oxygen and CO₂ by the blood. Regulation of breathing (Nervous and Chemical control).

2. Urinary System

Urine formation, composition of urine, urea clearance. Formation of concentrated and dilute urine, Regulation of osmolarity, and pH (acidic and basic urine), process of micturition (nervous control), renal failures.

3. Nervous System

Spinal reflexes. Reflex regulation of movement and posture. Cerebral cortex functions, Voluntary movements, Descending tracts of spinal cord. Basal ganglia, cerebellum, thalamus, C.S.F. Autonomic nervous system.

4. Special senses

Elementary knowledge of structure and function of the special senses.

5. Endocrinology

Definition of hormone. Nature of different types of hormone. Mechanism of action of hormones including pituitary hormones with abnormalities, thyroid gland with pathologies para thyroid hormone, pancreatic hormone with diabetes mellitus, Adrenal glands with cushing syndrome, Addison's disease. Male and female sex hormones.

6. Histology

Underlying principles of histological techniques and staining specific tissues. Staining of paraffin and frozen sections.

Books Recommended

1. Arthur C. Guyton and John E. Hall. Textbook of Medical Physiology. 13th Edition, 2015, W.B. Saunders, Philadelphia.
2. [Barbara Young](#), [Philip woodford](#), [Geraldine o' Dowd](#) [Wheater's Functional Histology: A Text and Colour Atlas](#). 6th Edition, 2013.
3. Douglas F. Paulsen. Basic Histology: Examination and Board Review. 5th edition, 2010, Prentice Hall Internal Inc.
4. Frederic H. Martini. Fundamentals of Anatomy and Physiology. 10th Edition, 2014, Prentice Hall, New Jersey.
5. Gerard J. Tortora & Bryan Derrickson. Principles of Anatomy and Physiology. 14th Edition, 2013, John Wiley & Sons, New York.

PHL-310

Pharmaceutical Biochemistry-II

Cr. Hrs. 3

1. Vitamins

Chemistry, classification (fat-soluble and water-soluble vitamins), biological and pharmaceutical importance of vitamins.

2. Hormones

Chemistry, classification (proteinous and non-proteinous hormones, amino acid derivatives, steroids), Biological and pharmaceutical importance of hormones.

3. Regulation of Metabolic Processes

Role of Vitamins, Physiological role of fat-soluble and water-soluble vitamins. Co-enzymes and their role in the regulation of metabolic processes. Niacin, thiamine, riboflavin, pyridoxine, pantothenic acid, biotin, folic acid and vitamin B₁₂.

4. Receptor Mediated Regulation (Hormones)

Mechanism of action of hormones, physiological roles of various hormones, site of synthesis and target sites of hormones, action, regulation, signal transduction mechanism, role of cAMP, calcium ions and phosphoinositides, tyrosine kinase, JAK-kinase in the regulation of metabolic processes.

5. Gene Expression

Regulation of gene expression, chemistry, transcription and translation, introduction to biotechnology and genetic engineering. Basic principles of recombinant DNA technology, Pharmaceutical applications. Genetic switch, inducers, fusion, genes, regulatory genes, zinc finger, Helix-turn helix motif, the leucine-zipper motif.

Books Recommended

1. [Alisa Peet, Michael A. Lieberman, Allan Marks. Marks' Basic Medical Biochemistry. 4th edition](#), 2012, Lippincott Williams & Wilkins.
2. [David L. Nelson, Michael M. Cox. Lehninger Principles of Biochemistry. 6th Edition, 2012, W.H. Freeman.](#)
3. MN Chatterjee & Rana Shinde. Text book of medical biochemistry. 8th edition, 2011 Jaypee Publishers.
4. Pamela C. Champe & Richard A. Harvey. Lippincott's Illustrated Review: Biochemistry. 6th Edition, 2013, Lippincott Williams & Wilkins, Philadelphia.
5. Robert K. Murray. Harper's Illustrated Biochemistry. 30th Edition, 2015, Lange Medical Books, McGraw Hill, New York.

PHL – 314

Anatomy

Cr. Hrs. 2

1. Introduction

Anatomical terminology, definition of cell, tissue, organ, structure of cell membrane, cytoplasm, organelles, nucleus, cell cycle.

2. **Tissues of Body**
Cartilage, bone structure and types of bones and joints.
3. **Muscle**
Structure of skeletal, smooth muscles, and cardiac muscles.
4. **Integumentary System**
Including skin, glands, hair and nail.
5. **Cardio Vascular System**
Structure of heart, location, blood supply to heart, types of blood vessels.
6. **Elementary System**
Name and structure of different parts of elementary system and their interrelationship.
7. **Urinary System**
Name and structure of organs of urinary system and their inter relationship.
8. **Male and Female Reproductive Systems**
Endocrine system including pituitary, thyroid and adrenal glands with their structures.
9. **Central Nervous System**
Including neuron, organization of CNS, brain, cerebrum, cerebellum, brain stem, Pons and medulla oblongata, thalamus, hypothalamus, cranial nerves. Internal structure of spinal cord CSF, sensory and motor pathways, spinal reflexes, peripheral spinal nerves.
10. **Autonomic Nervous System**
Sympathetic and Parasympathetic nervous system.

Books Recommended

1. [Edith Applegate](#). The Anatomy and Physiology Learning System. 4th edition, 2010, Saunders.
2. Frederic H. Martini. Fundamentals of Anatomy and Physiology. 10th Edition, 2014, Prentice Hall, New Jersey.
3. Gerard J. Tortora & Bryan Derrickson. Principles of Anatomy and Physiology. 14th Edition, 2013, John Wiley & Sons, New York.
4. [Richard L. Drake](#), [A. Wayne Vogl](#), [Adam W. M. Mitchell](#). [Gray's Anatomy for Students](#). 3rd edition, 2014, Churchill Livingstone.
5. [Kevin T. Patton](#), [Gary A. Thibodeau](#). Anatomy & Physiology . 8th edition, 2013, Mosby.

Second Professional

PHL-410

Systemic Pharmacology - I

Cr. Hrs. 3

1. **Autacoids and their Antagonists**

Serotonin and serotonin antagonist, other autacoids; Histamine and antihistamines, prostanooids.

2. Inflammatory Disorders

Medication for inflammatory disorders; Uricosuric agents; Glucocorticoids.

3. Drugs Acting on G.I.T. System

Gastroesophageal reflux disease and its treatment; emesis and antiemetics; Constipation and laxatives. Diarrhea and antidiarrheals; Peptic and duodenal ulcer, drugs used in their treatments; Drugs for inflammatory bowel disease, and for cholelithiasis; Prokinetics.

4. Chemotherapy

Introduction, principles of cell proliferation and chemotherapy. Principles of combination therapy, resistance of chemotherapy.

5. Chemotherapeutic Drugs

Antimicrobial, antiviral, antiprotozoal, antifungal, anthelmintic, antimycobacterial.

Books Recommended

1. [Anthony Trevor](#), [Bertram Katzung](#), [Susan Masters](#), [Marieke Knudering-Hall](#). Katzung & Trevor's Pharmacology Examination and Board Review, 14th Edition, 2015. Lange Medical Books.
2. Bertram G. Katzung, [Susan Masters](#), [Anthony Trevor](#). Basic and Clinical Pharmacology, 13th Edition, 2014. A Lange Medical Book. London.
3. David E. Golan, Armen H. Tashjian, Jr. Ehrin J. Armstrong, April W. Armstrong. Principles of Pharmacology "The Pathophysiologic Basis of Drug Therapy". 3rd Edition (2011). Lippincott Williams & Wilkins.
4. Goodman & Gilman's. The Pharmacological Basis of Therapeutics 12th Edition, 2010. McGraw-Hill, USA.
5. Rang H.P., Dale M.M. Rang & Dale's Pharmacology, 8th Edition, 2015. Churchill Living Stone, England.

Third Professional

PHL-510

Pharmacology (Lab)

Cr. Hrs. 3

1. Research methods and experimental techniques in pharmacology.
2. Development of experimental design and animal handling.
3. Routes of administration.
4. Preparation of physiological salt solutions.
5. To demonstrate the effects of sympathomimetic and sympatholytic drugs on frog's heart.
6. To demonstrate the effects of parasympathomimetic and parasympatholytic drugs on frog's heart.
7. To demonstrate the effects of an unknown drug on frog's heart
8. To demonstrate the effects of vasoconstrictor drugs on frog's blood vessels.
9. To demonstrate the effects of stimulant drugs on rabbit's intestine.
10. To demonstrate the effects of depressant drugs on rabbit's intestine.

11. To demonstrate the effects of an unknown drug on rabbit's intestine and identify the (unknown) drug.
12. To study the effects of adrenaline on rabbit's eyes.
13. To study the effects of homatropine on rabbit's eyes.
14. To study the effects of pilocarpine on rabbit's eyes.
15. To study the effects of local anesthetic drug.
16. To determine the analgesic response of the given drug.

Books Recommended

1. Bertram G. Katzung, [Susan Masters](#), [Anthony Trevor](#). Basic and Clinical Pharmacology, 13th Edition, 2014. A Lange Medical Book. London.
2. Bikash M, Ajay P. Practical Manual of Experimental and Clinical Pharmacology. 1st edition, 2010, Jaypee Brothers Medical Publishers.
3. [K.K. Pillai](#). Experimental Pharmacology. 2012, CBS Publisher.
4. Mark A. Suckow, Karla A. Stevens, Ronald P. Wilson. "The Laboratory Rabbit, Guinea Pig, Hamster, and Other Rodents". 1st edition, 2012, Academic Press.
5. Richard A. Harvey, Michelle A Clark, Richard Finkel, Jose A. Rey, Karen Whalen. Pharmacology (Lippincott's Illustrated Reviews). 6th edition, 2014, Lippincott Williams & Wilkins.

Fourth Professional

PHL-612

Pharmacology (Lab)

Cr. Hrs. 3

1. To study the convulsant effect of strychnine and picrotoxin in frogs and to determine the site of action.
2. To observe the effect of drugs on gross behavioral changes of animal.
3. To observe the effect of drugs on exploratory behavior of mice.
4. To identify the unknown (convulsant) drug and determine its site of action.
5. To observe the anti-inflammatory effect of given drug by hind paw method.
6. To observe the effect of a diuretic in animal.
7. To observe the hypoglycemic effect of drugs in rabbit.
8. To identify and differentiate the effects of unknown drug on human and the nerve plexus of frog.
9. To demonstrate the effects of acetylcholine on the rectus abdominus muscle of frog and competitive pharmacological antagonism by Neuromuscular blocking agents.
10. To identify the unknown drug by performing pharmacological competitive antagonism on rectus abdominus muscle of frog.
11. To study the effects of heparin and oral anticoagulants on rabbits.
12. To identify the unknown anticoagulant drug using rabbits.

13. To identify unknown concentration of acetylcholine from graded dose-response curves.
14. To observe the effect of drug on swimming induced depression.
15. To observe the effects of drug on learning behavior of animal.

Books Recommended

1. Bertram G. Katzung, [Susan Masters](#), [Anthony Trevor](#). Basic and Clinical Pharmacology, 13th Edition, 2014. A Lange Medical Book. London.
2. Bikash M, Ajay P. Practical Manual of Experimental and Clinical Pharmacology. 1st edition, 2010, Jaypee Brothers Medical Publishers.
3. [K.K.Pillai](#). Experimental Pharmacology. 2012, CBS Publisher.
4. [Mark A. Suckow](#), [Karla A. Stevens](#), [Ronald P. Wilson](#). "The Laboratory Rabbit, Guinea Pig, Hamster, and Other Rodents". 1st edition, 2012, Academic Press.
5. [Richard A. Harvey](#), [Michelle A Clark](#), [Richard Finkel](#), [Jose A. Rey](#) , [Karen Whalen](#). Pharmacology (Lippincott's Illustrated Reviews). 6th edition, 2014, Lippincott Williams & Wilkins.

Fifth Professional

PHL-712

Toxicology

Cr. Hrs. 2

1. **Principles of Toxicology**
Principles of treatment of poisoning, classification of toxic agents, spectrum of undesired effects, mechanisms of toxicity.
2. **Disposition of Toxicants**
Absorption, distribution and elimination of toxicants, biotransformation of xenobiotics.
3. **Target Organs of Toxicity**
Toxic responses of the blood, toxic responses of the liver, toxic responses of the heart, toxic responses of the kidney, toxic responses of reproductive system.
4. **Environmental Toxicology**
Air pollution and ecotoxicology.
5. **Applications of Toxicology**
Food toxicology, forensic toxicology, clinical toxicology, occupational toxicology, risk assessment.

Books Recommended

1. Bertram G. Katzung, [Susan Masters](#), [Anthony Trevor](#). Basic and Clinical Pharmacology, 13th Edition, 2014. A Lange Medical Book. London.

2. Curtis D. Klaassen. Casarett & Doull's Toxicology: The basic science of poisons. 8th Edition, 2013, McGraw Hill Medical Publishing Division, London.
3. Ernest Hodgson, Patricia E. Levi. A text book of modern toxicology. 4th edition. 2010. John Wiley & Sons. Inc.
4. Fank C. Lu and Sam Kacew. Lu's basic toxicology: Fundamentals, target organs and risk assessment, 6th Edition 2012, CRC Press
5. Timbrell J.A. Introduction to Toxicology. 3rd edition, 2001, Taylor & Francis Ltd.

Department of Pharmacognosy

Pharm-D Courses

First Year					
1st Semester			2nd Semester		
Course No.	Title of Course	Cr. Hr.	Course No.	Title of Course	Cr. Hr.
-	-		PHG-312	General Pharmacognosy	3
Second Year					
PHG-411	Pharmacognosy (Lab)-I	3	PHG-412	Chemical Pharmacognosy-I	3
Third Year					
PHG-511	Chemical Pharmacognosy-II	3	PHG-512	Pharmacognosy (Lab)-II	3
-	-		PHG-514	Natural Toxicants	2
Fourth Year					
PHG-611	Advance Pharmacognosy	3	-	-	-
Fifth Year					
PHG-713	Clinical Pharmacognosy	2	-	-	-
Total 08 Courses making 22 Credit Hours in five years.					

* English compulsory course is to be conducted in 1st year (Second Semester) as a non-credit course under the Department of Pharmacognosy.

FIRST SEMESTER

Second Professional

PHG-411

Pharmacognosy Lab-I

Cr. Hrs. 3

1. Macroscopic (Organoleptic) Examination of Natural Products

Organized Drugs

Over ground parts

Seed: *Melia azadirachta*, *Cucumis melo*, *Psoralea corylifolia*, *Lallementia royleana*, *Ricinus communis*, *Ipomea purga*, *Nigella sativa*, *Datura stramonium*, *Tamarix gallica*, *Strychnos nux-vomica*, *Punica granatum* and *Prunus amygdalus*.

Fruit: *Cuminum cyminum*, *Piper nigrum*, *Cubeba officinalis*, *Citrus limon*, *Embelica officinalis*, *Pongamia pinnata*, *Papaver somniferum*, *Aegle marmelos*, *Ficus carica*, *Terminalia chebula*, *Sphaeranthus indicus*, *Illicium verum*, *Tribulus terrestris*, *Prunus domestica*, *Gossypium herbacium* and *Phycotis ajowan*.

Pod: *Helicteres isora*, *Cassia fistula* and *Moringa olifera*.

Flower: *Punica granatum*, *Viola odorata*, *Hibiscus rosa-sinensis*, *Onosma bracteatum*.

Leaf: *Lawsonia innermis*, *Onosma bracteatum*, *Cinnamomum zeylanicum*, *Gymnema sylvestre*, *Catharanthus roseus*.

Bark: *Cinnamomum zeylanicum*, *Onosma echinoides*, *Valeriana hardwickii*.

Underground parts

Root and Rhizome: *Alpinia officinarum*, *Flacourtia indica*, *Curcuma amada*, *Zingiber officinale*, *Valeriana officinalis*, *Cuscuta reflexa*, *Glycyrrhiza glabra* and *Smilax china*.

Unorganized drugs

Extracts: *Aloe indica*, *Uncaria gambir*, *Papaver somniferum*.

Exudates: *Ferula foetida*, *Acacia arabica*, *Styrax benzoin*, *Bambusa arundinacea*, *Ipomea purga*, *Pinus longifolia*.

Animal drugs: *Bombyx mori*, *Laccifer lacca*, *Ovis aries*, *Cantheris vesicatoria*, *Cervus elephus*, *Apis melifera*.

Mineral drugs: Sulphur, Alum, Borax, Ammonium chloride, Asphalt.

The number of crude drugs may be increased on the availability of sources.

2. Microscopic Examination of Natural Products

Powdered Drug Examination

Bark: *Cinnamomum zeylanicum*.

Seed: *Strychnos nux-vomica*.

Flower bud: *Eugenia caryophyllus*.

Leaf: *Cassia senna*, *Lawsonia innermis*.

Fruit: *Coriandrum sativum*, *Elettaria cardamomum*, *Carum carvi*, *Hyocyamus niger*, *Myristica fragrans*, *Pimpinella anisum*, *Foeniculum vulgare*.

Root: *Glycyrrhiza glabra*, *Atropa belladonna*.

Rhizome: *Zingiber officinale*, *Rheum officinale*.

Resin: *Ipomea purga*, *Podophyllum* spp.

Histological Examination by Section Cutting.

Transverse sections of fresh and dried crude drugs

Stem: *Ephedra sinica*, *Hyocyamus niger*, *Atropa belladonna*, *Datura stramonium*, *Catharanthus roseus*.

Leaf: *Cassia senna*, *Mentha piperita*, *Digitalis purpurea*, *Eucalyptus globulus*.

Fruit: *Piper nigrum*, *Foeniculum vulgare*, *Coriandrum sativum*.

Seed: *Ricinus communis*.

Bark: *Cassia sagra*, *Cinnamomum zeylanicum*.

Root: *Aconitum* spp., *Rheum emodi*, *Atropa belladonna*, *Glycyrrhiza glabra* and *Rauwolfia serpentina*.

Rhizome: *Zingiber officinale*, *Curcuma longa*.

Flower bud: *Eugenia caryophyllus*.

3. Phytochemical Examination of Natural Products

Stem: *Cassia senna*, *Mentha piperita*, *Ephedra sinica*, *Hyoscyamus niger*, *Atropa belladonna*, *Datura stramonium*, *Catharanthus roseus*.

Root: *Aconitum* spp., *Rheum emodi*, *Atropa belladonna*, *Glycyrrhiza glabra*, *Rauwolfia serpentina*.

Rhizome: *Zingiber officinale*, *Podophyllum* spp., *Ipomoea purga*, *Rauwolfia serpentina*, *Rheum emodi*

Bark: *Cinchona* spp., *Cassia sagra*, *Cinnamomum zeylanicum*.

Leaves: *Mentha piperita*, *Digitalis purpurea*, *Cassia senna*, *Eucalyptus globulus*.

Fruit: *Foeniculum vulgare*, *Coriandrum sativum*.

Seed: *Strychnos nux-vomica*, *Myristica fragrans*, *Pimpinella anisum*, *Carum carvi*.

Flower: *Eugenia caryophyllus*.

4. Surgical Dressings

Examination of surgical dressings and test for dressings and sutures as given in B.P.C.

5. Analysis and Isolation of Natural Products

- Identification of fixed oils by colour reactions.
- Identification of tannins by colour reactions.
- Determination of the acid value of rosin.
- Determination of Iodine value of olive oil.
- Determination of volatile oil in Cinnamon.
- Determination of saponification value of cotton seed oil.
- Assay of benzoin.
- Assay of peppermint oil for total esters.
- Isolation of menthol from *Mentha piperita*.
- Isolation of rotenone from *Longicarpus* spp.
- Isolation of coumarin from *Melilotus officinalis*.
- Isolation of shikimic acid from *Illicium verum*.

Books Recommended

1. Practical Pharmacognosy, S.B Gokhale, Dr. C.K. Kokate, Ali Nirali Prakashan (2008).
2. Practical Pharmacognosy (9th ed.), K. R. Khandelwal, Ali Prakashan Nir, (2008).
3. Atlas of Microscopy of Medicinal Plant, Culinary Herbs and Spices, Betty P. Jackson, Derek W. Snowdown (Ed.), CBS Publishers and Distributors, (2005).
4. Encyclopedia of World Medicinal Plants (vol. 5), T. Pullaiah, Illustrated, Regency Publications, New Delhi (2006).

5. Quality Control of Herbal Drugs: An Approach to Evaluation of Botanicals (1st ed.), [Pulok K. Mukherjee](#) Publisher Business Horizons, New Delhi (2002).

Third Professional

PHG-511 Chemical Pharmacognosy-II Cr. Hrs. 3

1. Alkaloids

Introduction, biosynthesis, classification, extraction, isolation, pharmacological activity and pharmacognostic studies of Pyridine (*Nicotiana spp.*), Piperidine (*Nicotiana spp.*), Tropane (*Hyoscyamus, Atropa, Datura spp.*), Imidazole (*Pilocarpus spp.*), Isoquinoline (*Ipecac, Curare, Opium spp.*), Quinoline (*Cinchona spp.*), Indole (*Ergot, Rauwolfia, Catharanthus, Nux-vomica spp.*), Purine (*Thea spp.*), Steroidal (*Veratrum spp.*), Diterpene (*Aconitum spp.*), Phenethylamine (*Ephedra, Colchicum spp.*) alkaloids.

2. Glycosides

Introduction, classification, chemistry, extraction, isolation, pharmacology and medicinal uses of glycosides from *Glycyrrhiza glabra*, *Dioscorea spp.*, *Panax ginseng*, *Cascara sagrada*, *Cassia spp.*, *Aloe spp.*, *Rheum officinale*, and *Ruta graveolens*.

3. Steroids

Introduction, extraction, isolation, nomenclature, sources and uses of bile acids, plant sterols, cardiac glycosides, steroidal sapogenins, vitamins, steroidal hormones, withanolides and ecdysons.

4. Poisonous Plants

Introduction, sources and toxic manifestations of poisonous plants with special reference to Pakistan.

5. Herbal Practice

Study of traditional medical systems as Unnani, Ayurvedic and Homeopathic in Pakistan.

Books Recommended

1. Biosynthesis of Natural Products, Mannito P., John Wiley and Sons, New York, (1981).
2. Herbal Drugs and Phytopharmaceuticals, Nornam G. Bisset (Ed.), Medpharm Scientific Publishers, Stuttgart (1994).
3. Medicinal Natural Products: A Biosynthetic Approach (1st ed.), Paul M Dewick, John Wiley and Sons, Ltd., Chichester (1997).
4. Medicinal Plants of the World: Chemical Constituents, Traditional and Modern Medicinal Uses (vol. 1-3), Ivan A. Ross, Humana Press, New Jersey, (2005).
5. Poisonous Plants of All Countries (2nd ed.), Arthur-Bernhard Smith, Periodical Expert Book, Delhi (1988).

Fourth Professional

PHG-611 Advance Pharmacognosy Cr. Hrs. 3

1. Nutraceuticals

Introduction, classification, chemistry, role in healthcare system, toxicity, side effects, regulatory and market aspects of nutraceuticals. Role of flavonoids, isoflavones, lycopene, lignans, lutein, phytoestrogens, amino acids, lecithin, octacosanol, co-enzyme Q, vitamins, bromelain, flax seed oil, cranberry extract, tea extract, pycnogenol, ginkgobiloba, garlic, ginseng, flax, kelp, spirulina, chlorella, chitosan, glucosamine, chondroitin, creatine,

carnitine, bee pollen, shark cartilage, royal jelly, fish oil, superoxide dismutase, pro- and pre-biotics, fluoride, zinc, potassium, calcium and selenium as nutraceuticals.

2. Cosmeceuticals

Introduction, classification of cosmeceuticals as skin, eye, hair care and others products. Biological studies, chemistry, uses and toxicology of botanicals and their isolated products used in cosmeceuticals. Role of rosemary, lavender, arnica, chamomile, licorice, pomegranate, calendula, cinnamon, aloe, tea, carrot, jojoba, apricot, ginger, olives, coconut, turmeric, thyme, walnut, cucumber, henna, onion, sage, oats, hibiscus, propolis, bromelin, balsam, gum, myrrh, honey, kelp and Laminaria as cosmeceuticals.

3. Allergens and Allergenic Preparations

Introduction, classification of allergens, causes, history, skin test and treatment of allergies. Types of allergens: inhalent, ingestant, injectant, contactant, infectant and infestant.

4. Enzymes

Introduction, classification, chemistry and functions of plant, animal and microbial enzymes. Study of malt extract, pepsin, streptokinase and asparaginase.

5. Hormones

Introduction, historical background of uses, commercial production, extraction and isolation of animal hormones: adrenal gland, thyroid, pituitary, parathyroid, and pancreas; Plant hormones: auxins, gibberellin, cytokinins and abscisic acid.

6. Anticancer Agents

Introduction of anticancer agents of natural origin, as *Catharanthus roseus*, *Colchicum autumnale*, *Podophyllum peltatum*, rifamycin, antibiotics, macrolide antibiotics, Anti-AIDS agents and immunostimulants.

Books Recommended

1. Williams Textbook of Endocrinology (11th ed.), Henry Kronenberg, Shlomo Melmed, Kenneth Polonsky, Saunders, (2008).
2. Anticancer Agents from Natural Products, Gordon M. Cragg, David G. I. Kingston, David J. Newman, CRC Press (2005).
3. Nutraceuticals; A Guide for Healthcare Professionals (2nded.). Brain Lockwood, Pharmaceutical Press, London (2007).
4. Cosmeceuticals and Active Cosmetics: Drugs versus Cosmetics, Peter Elsner and Howard I. Maibach (Eds.), CRC Press (2005).
5. Plants in Cosmetics: Plants and Plant Preparations used as Ingredients for Cosmetic Products. Anton R., Patri F. Silano V., Council of Europe Publishing (2001).

Fifth Professional

PHG-713

Clinical Pharmacognosy

Cr. Hrs. 2

1. Introduction to Clinical Pharmacognosy

General introduction and historical background of clinical Pharmacognosy. Study of causes, pathogenesis, clinical features (sign and symptoms), diagnosis, prognosis, prevention and epidemiology of chronic diseases; principles of medication, treatment by herbal medicines.

2. Clinical Use of Herbal Medicine

Diabetes: *Gymnema sylvestre*, *Melia azadirchta*, *Momordica charantia*, *Syzygium jambulana*.
Cardiac diseases: *Digitalis spp.*, *Convallaria majalis*, *Urgenia indica*, *Allium sativum*, *Punica granatum*.

Hepatitis: *Berberis vulgaris*, *Picrorhiza kurroa*, *Lawsonia innermis*.

Respiratory diseases: *Ficus religiosa*, *Adhatoda vasica*.

Skin diseases: *Aloe vera*, *Angelica archangelica*, *Mentha piperita*, *Citrus spp.*, *Commiphora mukul*.

CNS disorders: *Strychnos nux-vomica*, *Datura stramonium*, *Cannabis sativa*, *Papaver somniferum*, *Atropa belladonna*.

Musculo-skeletal disorders: *Nigella sativa*, *Phycotis ajowan*, *Trigonella foenum-graecum*, *Zingiber officinale*.

Renal disorders: *Cucumis melo*, *Berberis vulgaris*, *Zea mays*, *Tribulus terrestris*.

Reproductive disorders: *Saraca indica*, *Ruta graveolens*, *Nigella sativa*, *Glycyrrhiza glabra*, *Claviceps purpurea*, *Myristica fragrance*.

G.I.T. disorders: *Foeniculum vulgare*, *Ferula foetida*, *Cuminum cyminum*, *Aegle marmelos*, *Prunus domestica*.

Books Recommended

1. Principle and Practice of Phytotherapy: Modern Herbal Medicine, Kerry Bone, Simons Mills, Churchill Livingstone (2013).
2. Pharmacological Methods in Phytotherapy Research, Selection, Preparation and Pharmacological Evaluation of Plant Material (vol.1), Elizabeth M. Williamson, David T. Okpako, Fred J. Evans, Wiley Publishers, (1996)
3. Encyclopedia of World Medicinal Plants (vol. 5), T-Pullaiah., Regency Publication, New Delhi (2006).
4. Natural Products: Drug Discovery and Therapeutic Medicine, Lixin Zhang, Arnold L. Demain (Eds.), Humana Press (2005).
5. Handbook of Clinical Pharmacy, Dr. A. V. Yadav, B.V. Yadav, T. I. Shaikh, Nirali Parkashan, Pune (2008).

SECOND SEMESTER

First Professional

PHG-312

General Pharmacognosy

Cr. Hrs. 3

1. Historical Development and Scope of Pharmacognosy

General introduction, history of Pharmacognosy with special reference to the contribution of Muslim scientists to the profession of Pharmacy. Introduction to herbal pharmacopoeias and modern concepts about Pharmacognosy.

2. Crude Drugs

Crude drugs, commerce, preparation, chemical and therapeutic classifications of crude drugs; official and un-official drugs.

3. Evaluation and Adulteration of Crude Drugs

Organoleptic study, physical evaluation, microscopic evaluation, types of adulteration, inferiority, spoilage, admixture, sophistication and substitution of crude drugs.

4. Drugs of Animal Origin

General introduction and discussion about honey, gelatin, shellac, musk, civet, ambergris, cod liver oil, cantharides and spermaceti.

5. Biologics

Sources, structure, preparation, description and uses of vaccines, toxins, antitoxins, venoms, antivenoms, antiserums.

6. Surgical Dressings

Classification of fibers as vegetable, animals and synthetic fibers. Evaluation of fibers in surgical dressings, BPC standards for dressings and sutures. Discussion on cotton, wool, cellulose, rayon, catgut and nylon.

7. Pesticides

Introduction, methods and control of pests with special reference to pyrethrum, tobacco, and other natural pesticides.

Books Recommended

1. Trease and Evans' Pharmacognosy (16th ed.), William C. Evans, George E. Trease, Daphne Evans, Elsevier (2009).
2. Pharmacognosy (9th ed.), Tyler V.E., Brady L.R. and Robbers J.E., Lea and Febiger, Philadelphia (1988).
3. Biologics in General Medicine, Wolf- Henning Boehncke, Heinfried H. Radeke, Springer, (2007).
4. Quality Control of Herbal Drugs: An Approach to Evaluation of Botanicals (1st ed.), Pulk K. Mukherjee, Business Horizons (2002).
5. Dictionary of Natural Products: Terms in the Field of Pharmacognosy Relating to Natural Medicinal and Pharmaceutical Materials and the Plants, Animal and Minerals from Which They are Derived (2nd ed.), George MacDonald Hocking, Plexus Publication, (1997).

Second Professional

PHG-412

Chemical Pharmacognosy-I

C. Hrs. 3

1. Carbohydrates and Related Compounds

Introduction and classification of carbohydrates, sugars as adjuvant in drugs, role of impurities in sugar substances.

2. Fixed Oils

Introduction, classification, source, active constituents and pharmacological uses of castor oil, cotton seed oil, olive oil, peanut oil, sesame oil, sunflower oil, chaulmoogra oil, corn oil, coconut oil, almond oil, linseed oil, mustard oil, palm oil and soya oil.

3. Essential Oils

Introduction, classification, sources, active constituents and pharmacological uses of clove oil, fennel oil, coriander oil, orange oil, eucalyptus oil, peppermint oil, anise oil, chenopodium oil, turpentine oil, lemon peel oil, citronella oil, caraway oil and thyme oil

4. Resins and Oleoresins

Introduction, classification, active constituents, pharmacological uses of jalap, turpentine, asafoetida, benzoin, rosin, cannabis, podophyllum, ipomea, myrrh, and balsam.

5. Tannins

Introduction, classification, biosynthesis, extraction, identification, occurrence in plants, their role in plant life and chemical study of tannins in kino, myroblan, catechu, nutgall, castanea, and krameria.

6. Chromatography

Introduction, types and application of chromatography for the separation, isolation, purification and identification of drug substances from natural sources.

Books Recommended

1. Biosynthesis of Natural Products, Manitto P., John Wiley and Sons, New York (1981).
2. Chromatography: Fundamentals and Applications of Chromatographic and Electrophoretic Methods. Part A, Fundamentals and Techniques, E. Heftmann, Elsevier Scientific Publishing Company, Amsterdam (1983).
3. Herbal Drugs and Phytopharmaceuticals, Nornam G. Bisset (Ed.), Medpharm Scientific Publishers, Stuttgart (1994).
4. Medicinal Natural Products: A Biosynthetic Approach (1st ed.), Paul M Dewick, John Wiley and Sons, Ltd., Chichester (1997).
5. Medicinal Plants of the World: Chemical Constituents, Traditional and Modern Medicinal Uses (vol. 1-3), Ivan A. Ross, Humana Press, New Jersey (2005).

Third Professional**PHG-512****Pharmacognosy Lab-II****Cr. Hrs. 3****1. Preliminary Screening of Natural Products**

Preliminary chemical tests for the detection of carbohydrates, tannins, alkaloids, glycosides, steroids, saponins, terpenes and flavonoids.

- Alkaloids: Mayer's reagent test, Wagner's reagent test, Dragendorff's reagent test.
- Glycosides: Kedde reagent test, Keller killiani test.
- Saponins: Haemolysis test, Froth test, Leibermann-Burchard test.
- Sterols: Salkowski test.
- Flavonoids: Colour test, Cyanidin test.
- Tannins: Ferric chlorides test, Lead acetate test.

2. Chromatography

- To show the presence of strychnine and brucine in *Strychnos nux-vomica* mixture (acid or alkaline) by thin layer chromatography (TLC).
- To examine tinctures or extracts of *Digitalis purpurea* leaf by paper chromatography.
- To separate different constituents of alkaloid mixture or colouring substances by column chromatography.
- To determine ephedrine hydrochloride through ion exchange chromatography.
- To separate alkaloids from an extract or tincture of *Atropa belladonna* through ion exchange chromatography.

3. Isolation and Identification of Natural Products

- Hyoscyamine from *Atropa belladonna*.
- Piperine from *Piper nigrum*.
- Strychnine, brucine from *Strychnos nux-vomica*.
- Caffeine from *Thea sinensis*.
- Jalap resin from *Ipomea purga*.
- Starch from *Solanum tuberosum*.
- Myristicin and trimyristicin from *Myristica fragrans*.
- Pectin from *Citrus paradisi*.
- Extraction of Clove oil by steam distillation.
- Identification of fixed oil by colour reactions.

4. Spectroscopy

- Interpretation of I.R. spectra of antibiotics and glycosides.
- Interpretation of U.V. spectra of indole-alkaloids.
- Separation and identification of essential oils by GC-MS.

Books Recommended

1. Chromatography: Fundamentals and Applications of Chromatographic and Electrophoretic Methods. Part A, Fundamentals and Techniques, E. Heftmann, Elsevier Scientific Publishing Company, Amsterdam (1983).
2. Clarke's Isolation and Identification of Drugs, Moffat A. C., Pharmaceutical Press, London (1975).
3. Plant Drug Analysis: A Thin Layer Chromatography Atlas (2nd ed.), Hildebert Wagner, Sabine Bladt, Springer-Verlag, New York (1996).
4. Principles of Instrumental Analysis (5th ed.), Douglas A. Skoog, F. James Holler, Timothy A. Nieman, Saunders College Publishing, Philadelphia (1998).
5. Thin-Layer Chromatography: A Laboratory Handbook, Egon Stahl, Springer-Verlag (1969).

Third Professional**PHG-514****Natural Toxicants****Cr. Hrs. 2****1. General Introduction to Natural Toxicants**

Definition, classification and chemical nature of natural toxins and toxicities in humans and animals.

2. Higher Plant Toxins

Essential oils: Terpene (cineol, pine oil), Phenylpropane (apiol, safrole, myristicin), Monoterpene (thujone, menthafuran) Plant acids (oxalic acid, amino acid, resin acid), Glycosides (cardiotonic, cyanogenic), Alkaloids (imidazole, pyrrolizidine, tropane).

3. Lower Plant Toxins

Bacterial toxins (*Staphylococcus aureus*, *Clostridium botulinum*), Algal toxins (*Microcystis aeruginosa*, Cyanobacteria, *Gonyaulax cantenella*).

4. Mycotoxins

Fungal toxins (*Aspergillus* spp., *Claviceps purpurea*), Mushrooms (*Amanita* spp.).

5. Study of Toxins, their Prevention and Control Methods

Description, pharmacognostic features, pharmacological actions, chemical constituents, treatment, side-effects, contra-indications, warnings, prevention and control methods of *Abrus precatorius*, *Papaver somniferum*, *Eucalyptus* spp., *Nicotiana tabacum*, *Cannabis sativa*, *Digitalis purpurea*, *Datura stramonium* poisoning.

Books Recommended

1. Dictionary of Plant Toxins, Jeffery B. Harborne, Herbert Baxter, Gerard P. Moss (Eds.), Chichester, John Wiley (1996).
2. Herbal Products, Toxicology and Clinical Pharmacology (2nd ed.), Timothy S. Tracy, Richard L. Kingston (Eds.), Humana Press, Totowa (2007).
3. Poisonous Plants of Pakistan, S. H. Abid Askari, Oxford University Press, 2010.
4. Poisonous Plants: A Hand Book for Doctors, Pharmacists, Toxicologists, Biologists and Veterinarians (2nd ed.), Dietrich Forhne, Hans J Pfander, Manson Publishing, London (2004).
5. Poisonous Plants and Related Toxins, Thomas Acamovic, Colin S. Stewart, T. W. Pennycott, CABI (2003).

English Mandatory

Non-Credit

Course

1. Parts of speech and articles
2. Verb – Transitive and intransitive verbs
3. Active and passive voice
4. Modal verbs
5. Skimming and scanning
6. Precise writing and comprehension
7. Letter writing
8. Memo
9. Minutes of the meeting
10. How to write research proposals + term papers
11. Presentation skills
12. Technical writing
13. Note – taking
14. Essay writing

Books Recommended

1. English Grammar in use by Raymond Murphy
2. The Art of Public speaking by Stephen E. Lucas
3. The North Field Guide to writing, with Readings Editions by Richard Bullock Maureen
4. Dalay Goggin
5. Critical Theory Today: A user-friendly guide by Lois Tyson
6. A Practical English Grammar by A. J. Thomson & A. V. Martinet
7. Practical English Usage by Michael Swan
8. The Power of Communication: Skills to build trust, inspire loyalty and lead effectively by Helio Fred Gracia.

9. Power Listening: Mastering the most critical business skills of all by Bernard T. Ferrari.
10. English Grammar by Wren & Martin

Department of Pharmacy Practice

Pharm-D Courses

1st Semester			2nd Semester		
Course No.	Title of Course	Cr. Hr.	Course No.	Title of Course	Cr. Hr.
Fourth Year					
PHP-601	Pharmacy Practice- Hospital Pharmacy	3	PHP-602	Pharmacy Practice- Community, Social and Administrative Pharmacy	<u>3</u>
Fifth Year					
PHP-705	Pharmacy Practice- Clinical Pharmacy	3	PHP-714	Pharmacy Practice- Pharmacotherapeutics	<u>3</u>
Total 04 Courses making 12 Credit Hours in five years.					

FIRST SEMESTER

Fourth Professional

PHP-601 Pharmacy Practice: Hospital Pharmacy Cr. Hrs. 3

1. Hospital and Its Organization:
 - a. Brief history of hospital
 - b. Classification of Hospitals
 - c. Minimum requirements for registering a hospitals
 - d. Organizational Structure
 - e. Administration
 - f. Clinical Departments
 - g. Nursing, Dietetic, Pathology, Blood Bank, Radiology and other supportive services
 - h. Hospital Finances
2. Role of Pharmacy in Hospitals, role of pharmacist in small hospitals, nursing homes etc.
3. Minimum Standards for Hospital Pharmacies as per ASHP
 - Pharmacy, Its Organization and Personnel:
 - a. The Physical plant and its equipment:
 - b. Pharmacy specialist
 - c. Drug information Centre
 - d. Poison Control Centre and Antidote Bank
 - e. Pharmacy Education, CEs and CMEs, Internship program, clerkship, etc.
 - f. Professional services rendered
 - g. The Pharmacy: Central Sterile Supply Room (Eye drops, Ear drops) and Hyperalimentation.
 - h. Nuclear Pharmacy
 - i. Investigational use of drugs:
 - j. Health accessories, Ancillary supplies and Surgical supplies
4. Dispensing to Inpatients:
 - a. Methods of Dispensing & SOP's
 - b. Unit dose dispensing
 - c. Other concepts of dispensing, Satellite Pharmacy etc.
5. Dispensing To Ambulatory Patients
6. The Hospital Formulary System: guidelines of AHFS
7. Controlled Substances: Procurement, Distribution, dispensation and inventory management
8. Dispensing during off-Hours, off site Pharmacies
9. Purchasing, Distribution and Control of Hospital Medicines, Medical & surgical supplies
Supply Chain Managements and Execution: Purchasing, Stocking, Stock Control, Inventory Management, Drug Distribution, Relationship between purchasing, Distribution and Clinical Pharmacy Services
10. 200 contact hours Hospital Pharmacy internship in a tertiary care general hospital covering the clinical units.

Books Recommended

1. Pharmacy Practice By Kevin M. G. Taylor, Geoffrey Harding; CRC Press
2. Holdford, D. A. and T. R. Brown (2010). Introduction to Hospital and Health-System Pharmacy Practice, ASHP.
3. Thompson, J. E. (2009). A practical guide to contemporary pharmacy practice Third Edition, Williams & Wilkins Baltimore (MD).
4. Brown, T. R. (2006). Handbook of institutional pharmacy practice, ASHP.
5. Stephens, M. (2011). Hospital pharmacy, Pharmaceutical Press.
6. Hospital Pharmacy by W.E. Hassan, Lea & Fibiger

Fifth Professional

PHP-705

Pharmacy Practice: Clinical Pharmacy

Cr. Hrs. 3

1. Rationalize use of drugs
2. Medication Errors
3. Drug Utilization and Drug Utilization Review
4. Lab data interpretation and monitoring (ADR etc.)
5. Developing and implementing TDM.
6. Documentation of Pharmacotherapy Plan
 - SOAP (Subjective, Objective, Assessment and Plan) note
 - CORE (Communication, Organization, Relationship and Expectations) Pharmacotherapy Plan
 - PRIME (Pharmaceutical based problems, Risk to patient, Interactions, Mismatch between medications and conditions or patients need and Efficacy issues) Pharmacotherapy problems
 - FARM (Findings, Assessment, Resolution and Monitoring and Follow up) note
7. Pharmacotherapy Decision making.
8. Pharmaco-economics and its implementation
9. Public health issues and different models to study e.g tobacco cessation programs, obesity prevention, tuberculosis monitoring, vaccinations etc.
10. CNS Unit (Psychosis, Stroke, Epilepsy), GI Unit
11. Oncology Unit (Solid and other cancers)
12. Osso Unit (Bone Density, rickets, osteoporosis)
13. Gastroenterology Unit (Ulcers, cirrhosis, hepatitis etc.)

Books Recommended

1. Pharmacotherapy A Pathophysiologic Approach 9/E By Joseph DiPiro, Robert L. Talbert, Gary Yee, Barbara Wells, L. Michael Posey; McGraw Hill Professional
2. Thompson, J. E. (2009). A practical guide to contemporary pharmacy practice Third Edition, Williams & Wilkins Baltimore (MD).
3. Lauster, C. D. and S. B. Srivastava (2013). Fundamental Skills for Patient Care in Pharmacy Practice, Jones & Bartlett Publishers.
4. Lexi-Comp, Inc. OnlineTM, L.-C., L.-D. OnlineTM, et al. (2015).

5. Ansel, H. C. (2012). Pharmaceutical calculations, Lippincott Williams & Wilkins
6. Koda-Kimble, M. A., B. K. Alldredge, et al. (2012). Koda-Kimble and Young's Applied Therapeutics: The Clinical Use of Drugs, Lippincott Williams & Wilkins.
7. Chisholm-Burns, M. A., B. G. Wells, et al. (2010). Pharmacotherapy principles & practice, McGraw-Hill New York.

SECOND SEMESTER

Fourth Professional

PHP-602

**Pharmacy Practice: Community, Social &
Administrative Pharmacy**

Cr. Hrs. 3

1. Definitions and background: Health and its types, Fundamentals of health, Health determinants, Health Care System, Health care Organizations and Institutes, Public health and community pharmacy:
2. Epidemiology, Epidemiological methodology with a focus on specific disease states, pharmacoepidemiology.
3. Preventive health (EPI & CDC), family planning and Health policy.
4. Medical complication of drug taking: general and socio-economic Aspects.
5. Pharmacy layout design: objectives of layout design, types of community Pharmacies (Pharmaceutical center, prescription-oriented pharmacies, traditional pharmacies and the super drug stores Chain of Pharmacies), Consumer goods and purchases, classes of layout designs, Principles and characteristics of layout design and traffic flow analysis.
6. Operations in a Community Pharmacy
 1. Procurement of Drugs, Medical devices, Health accessories, ancillary supplies
 2. Good Storage Practices in community Pharmacy
 3. Compounding and dispensing including:
Fundamental operations in compounding, containers and closures for dispensed products,
 4. Prescription-handling (parts of prescription, filling, interpretation, pricing) and labelling of dispensed medication.
 5. Extemporaneous dispensing: solutions, suspensions, emulsions, creams, Ointments, pastes and gels, suppositories and pessaries, powders and granules and oral unit dosage form.
 6. Pharmaceutical incompatibilities: types of incompatibilities, manifestations, correction and prevention with reference to typical examples.
 7. Patient education and counselling, educational role of community pharmacy
 8. Control of drug abuse and misuse in community pharmacy
7. Role of pharmacist: as public health educator in the community for drug Monitoring and drug information.
8. Pharmacoeconomics: pharmacoeconomic modelling and interpretation.
9. Alternative therapies: background, philosophy and use of complementary and Alternative therapies in practice including herbal medicines, homoeopathy, acupuncture, acupressure, Bach Flower remedies, aromatherapy and reflexology.
10. One month training in Community Pharmacy and DIC under the supervision of licensed registered Pharmacist

Books Recommended

1. Jean-Venable R. Goode, Lynne M. Roman, Kristin W. Weitzel. Community Pharmacy Practice Case Studies
2. Pharmacy Practice By Kevin M. G. Taylor, Geoffrey Harding; CRC Press
3. Pharmacoeconomics: From Theory to Practice By Renee J. G. Arnold
4. Community Pharmacy By Paul Rutter
5. Thompson, J. E. (2009). A practical guide to contemporary pharmacy practice Third Edition, Williams & Wilkins Baltimore (MD).
6. Lauster, C. D. and S. B. Srivastava (2013). Fundamental Skills for Patient Care in Pharmacy Practice, Jones & Bartlett Publishers.
7. Beardsley, R. S., C. L. Kimberlin, et al. (2012). Communication skills in pharmacy practice: a practical guide for students and practitioners, Lippincott Williams & Wilkins.
8. Leadership and Management in Pharmacy Practice, Second Edition (Pharmacy Education Series). Jan 2, 2015. CRC Press

Fifth Professional

PHP-714 Pharmacy Practice: Pharmacotherapeutics Cr. Hrs. 3

Pharmacotherapy of drugs used in:

1. CVS Unit (covering IHD, Hypertension etc.).
2. Pulmonary Unit (Asthma, COPD, pneumonia etc.).
3. Infection Unit (Bacterial, Fungal and viral infections).
4. Endocrinology Unit (DM, Hyper / Hypothyroidism etc.).
5. Nephrology Unit (Renal failures etc.).
6. Hematology Unit (bleeding disorders, coagulopathies)
7. Manufacturing Bulk and Sterile: Total Parenteral Nutrition (TPN)

Books Recommended

1. Pharmacotherapy A Pathophysiologic Approach 9/E By Joseph DiPiro, Robert L. Talbert, Gary Yee, Barbara Wells, L. Michael Posey; McGraw Hill Professional
2. Lauster, C. D. and S. B. Srivastava (2013). Fundamental Skills for Patient Care in Pharmacy Practice, Jones & Bartlett Publishers.
3. Lexi-Comp, Inc. OnlineTM, L.-C., L.-D. OnlineTM, et al. (2015).
4. Koda-Kimble, M. A., B. K. Alldredge, et al. (2012). Koda-Kimble and Young's Applied Therapeutics: The Clinical Use of Drugs, Lippincott Williams & Wilkins.
5. Chisholm-Burns, M. A., B. G. Wells, et al. (2010). Pharmacotherapy principles & practice, McGraw-Hill New York.